SLA OR NAY?

CAUSAL IMPACTS OF NEIGHBORHOOD-LEVEL RENT QUALITY REGULATION

Levi John Wolf & Alex Marsh
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WHAT & WHY ARE DLA?

IDENTIFICATION & ESTIMATION

PRELIMINARY RESULTS

SLA OR NAY?

WHAT & WHY ARE DLA?

IDENTIFICATION & ESTIMATION

PRELIMINARY RESULTS

SLA OR NAY?

DISCRETIONARY LICENSING AUTHORITY

requirements for privately rented properties falling within a council's jurisdiction that extend national requirements for licensing

DISCRETIONARY LICENSING AUTHORITY

ADDITIONAL LICENSING

extends national HMO requirements to "smaller" households of 3 or more people w/ shared facilities

DISCRETIONARY LICENSING

DISCRETIONARY LICENSING AUTHORITY

extends national HMO requirements to all properties for 5 years

ADDITIONAL LICENSING

SELECTIVE LICENSING

DISCRETIONARY LICENSING

DISCRETIONARY LICENSING AUTHORITY

applies

ADDITIONAL LICENSING

to certain kinds of properties

SELECTIVE LICENSING

in specific areas & times

DISCRETIONARY LICENSING

EASTERN SLA (1)

Addressing severe quality issues
2016-2021 in St. George West & Eastville
All privately-let properties
EPC, Electric, Gas safety inspections
£300 to £750 fee

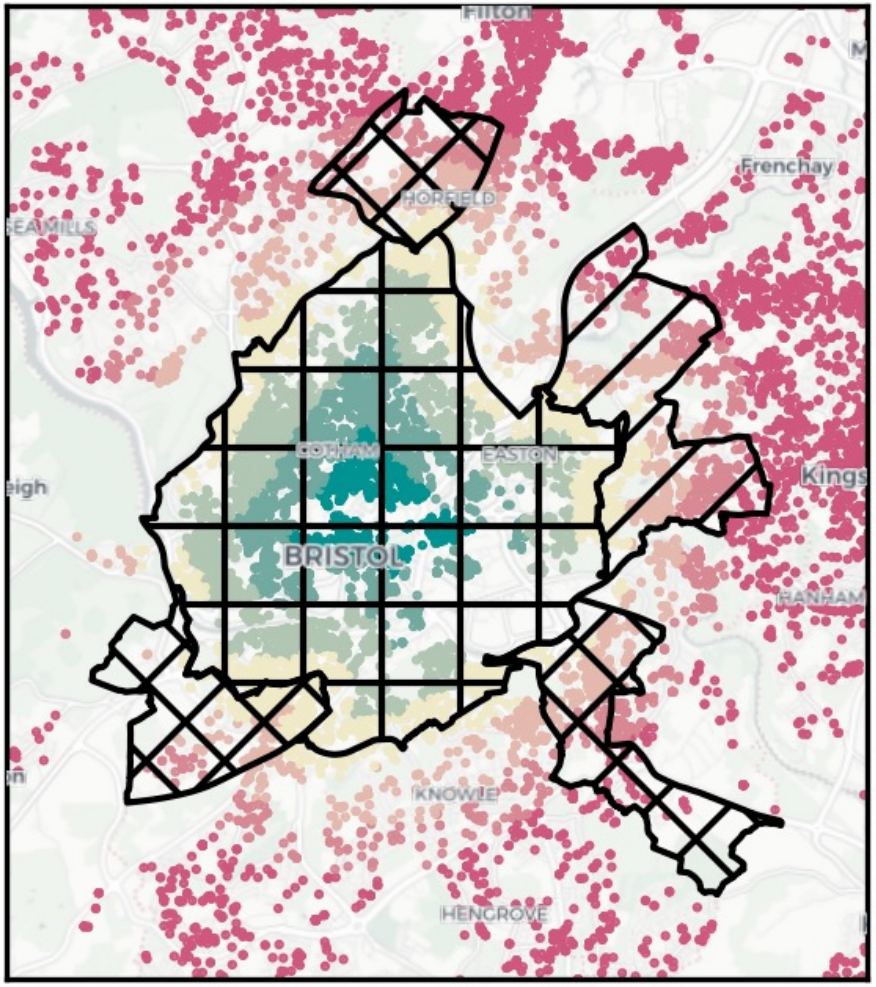
CENTRAL ALA (2)

2019-2024 in all central wards
3+ People w/ Shared Facilities
EPC, Electric, Gas safety inspections
£900 to £1450 fee

EXTENSION (3)

2024-2029 for different targets
EPC, Electric, Gas safety inspections
£900 to £2000 fee

Listings near DLAs



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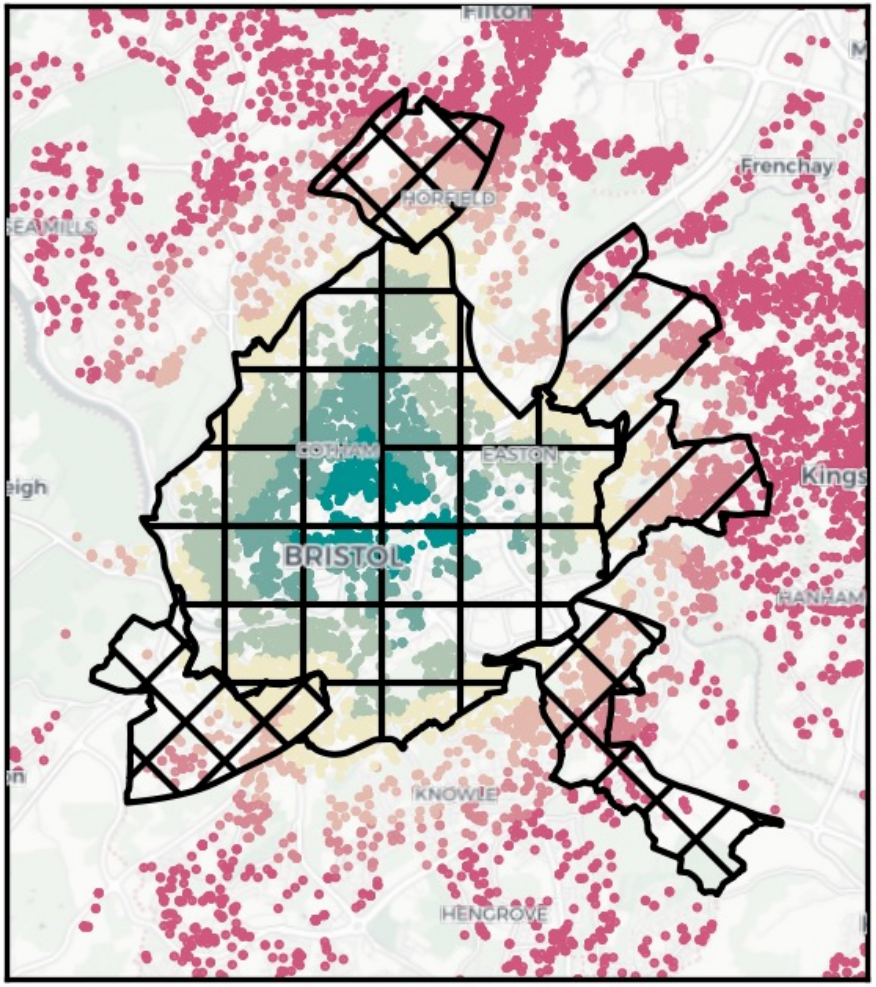
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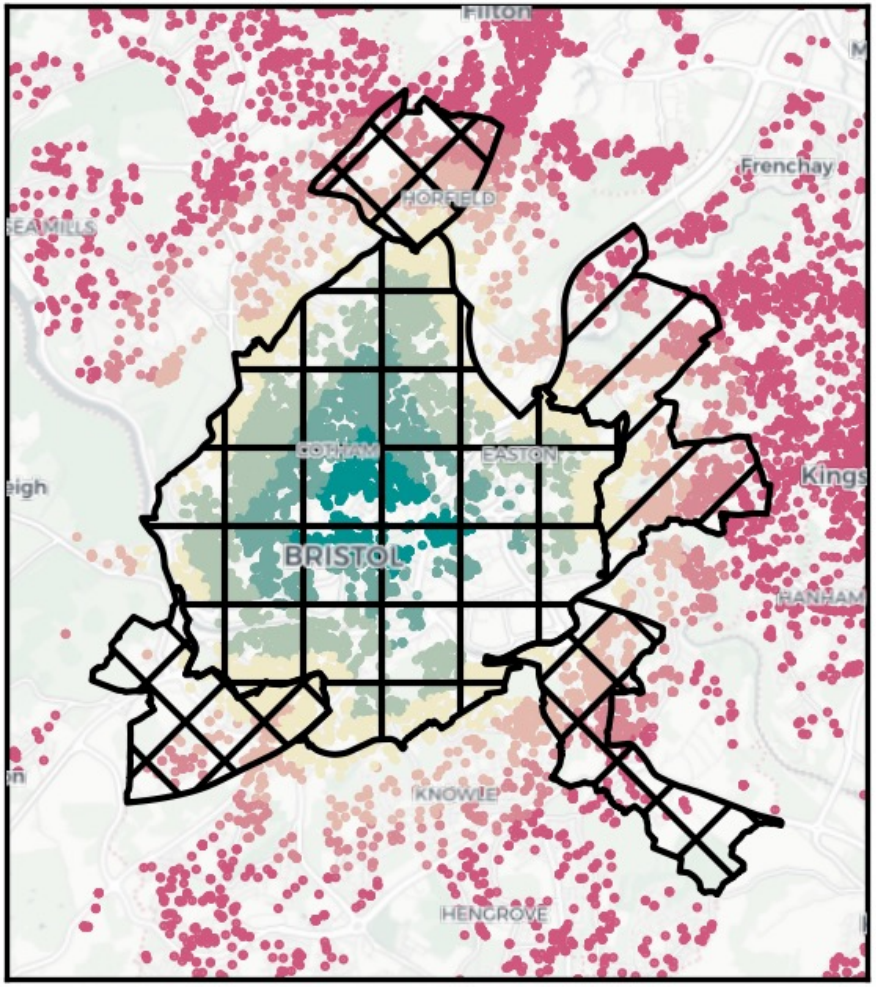
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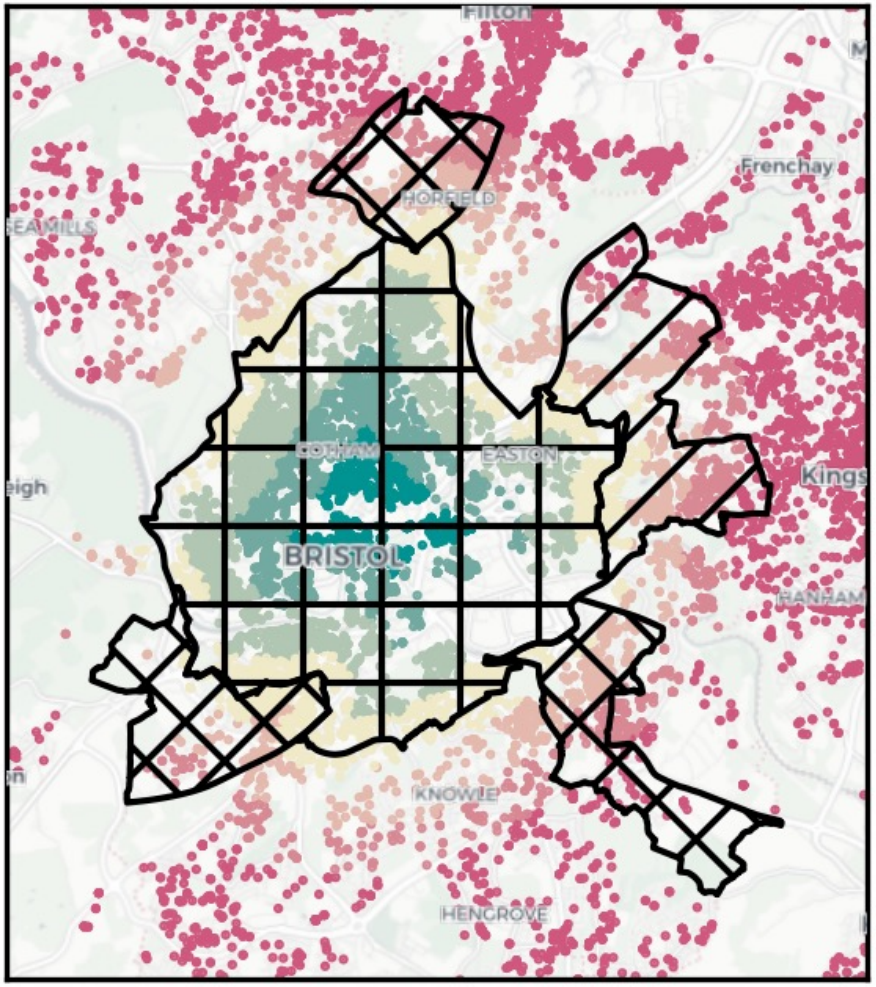
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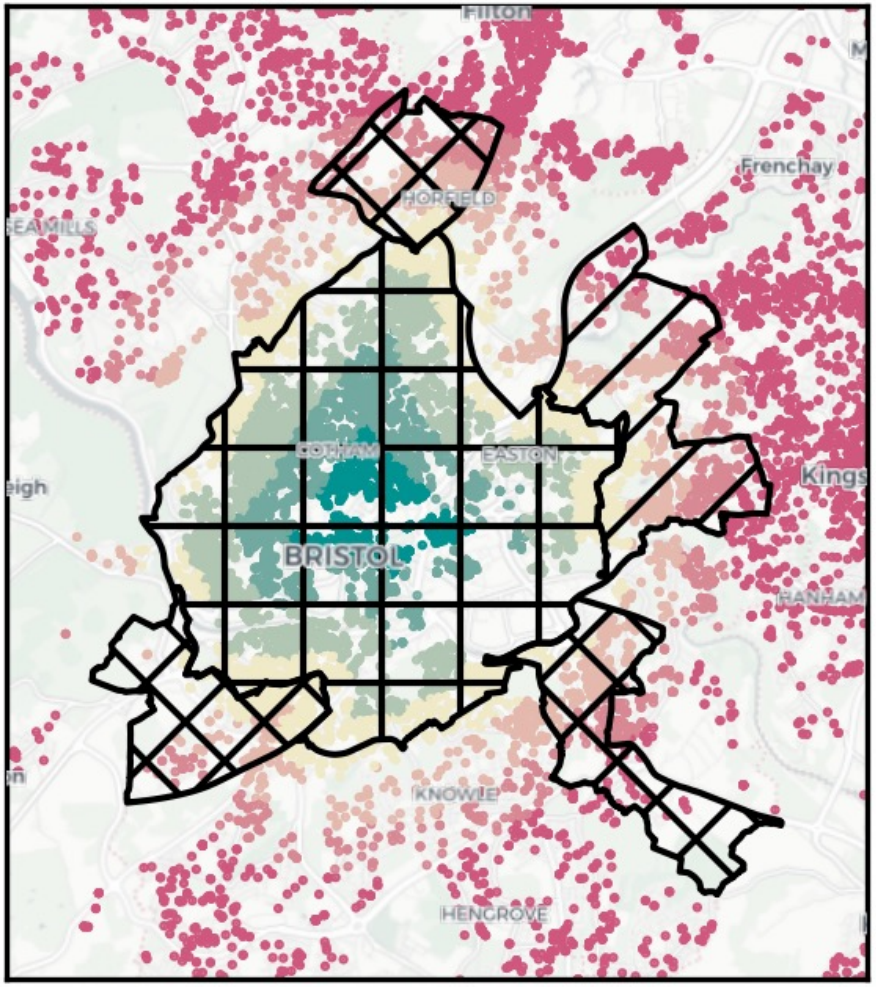
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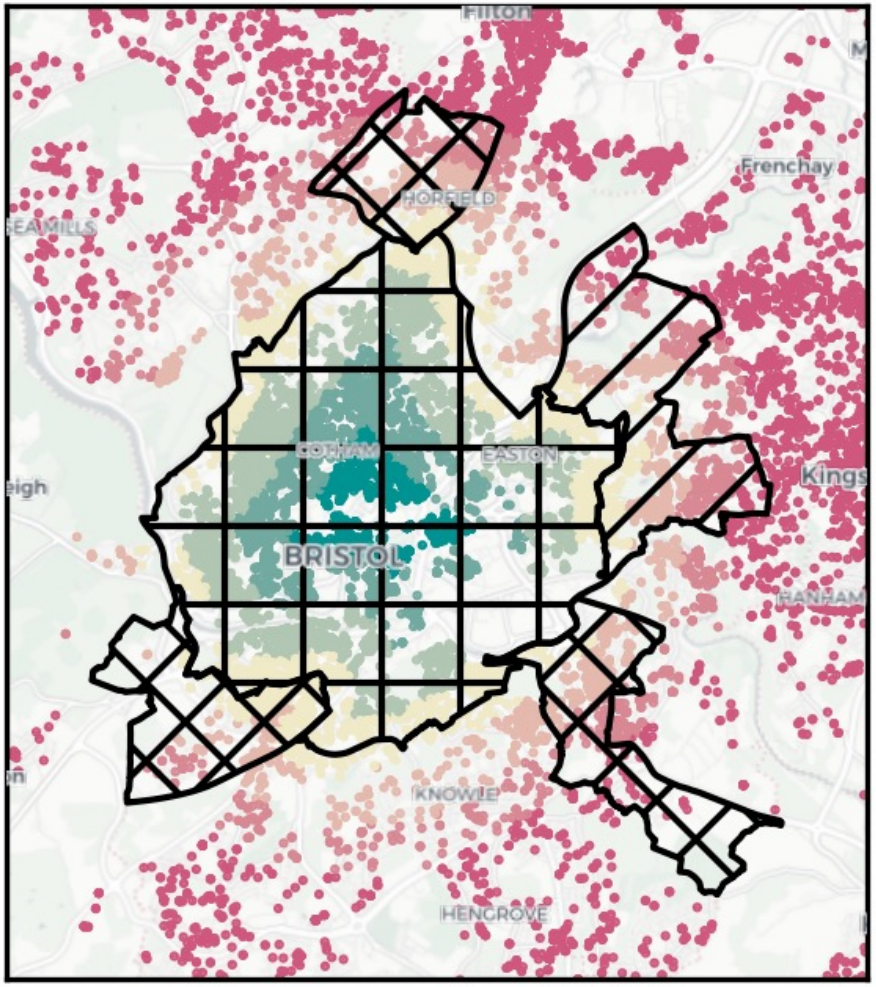
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Listings near DLAs



City Council of Bristol (The council)
Housing Act 2004 (“The Act”): Parts 2 and 3
NOTICE OF DESIGNATION OF A CITYWIDE ADDITIONAL LICENSING
NOTICE OF DESIGNATION OF BISHOPSTON AND ASHLEY DOWN, COTHAM AND EASTON WARDS FOR SELECTIVE LICENSING



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BBC

Bristol landlords oppose new licensing fee scheme

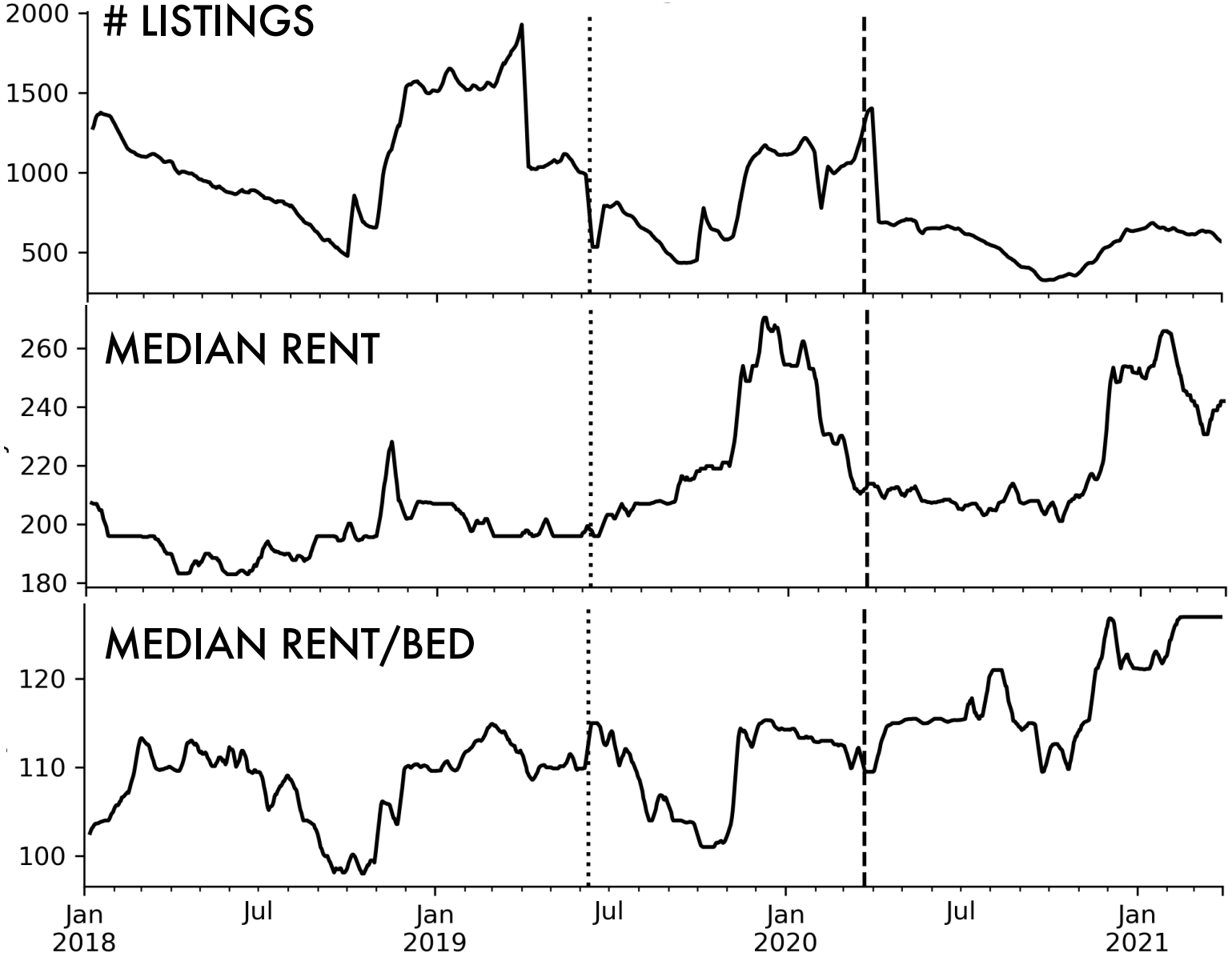
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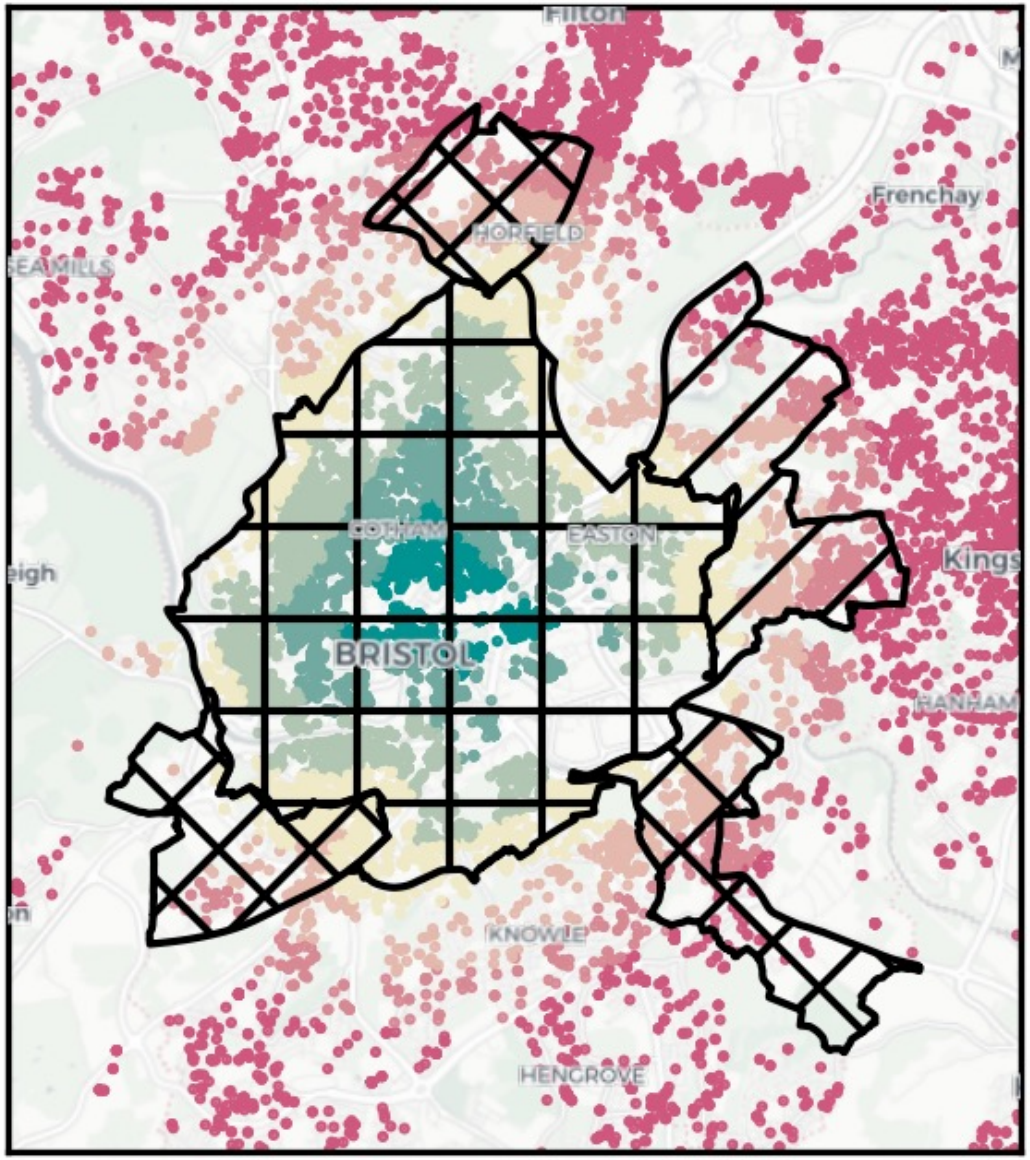
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Fewer rentals?

Higher rent?

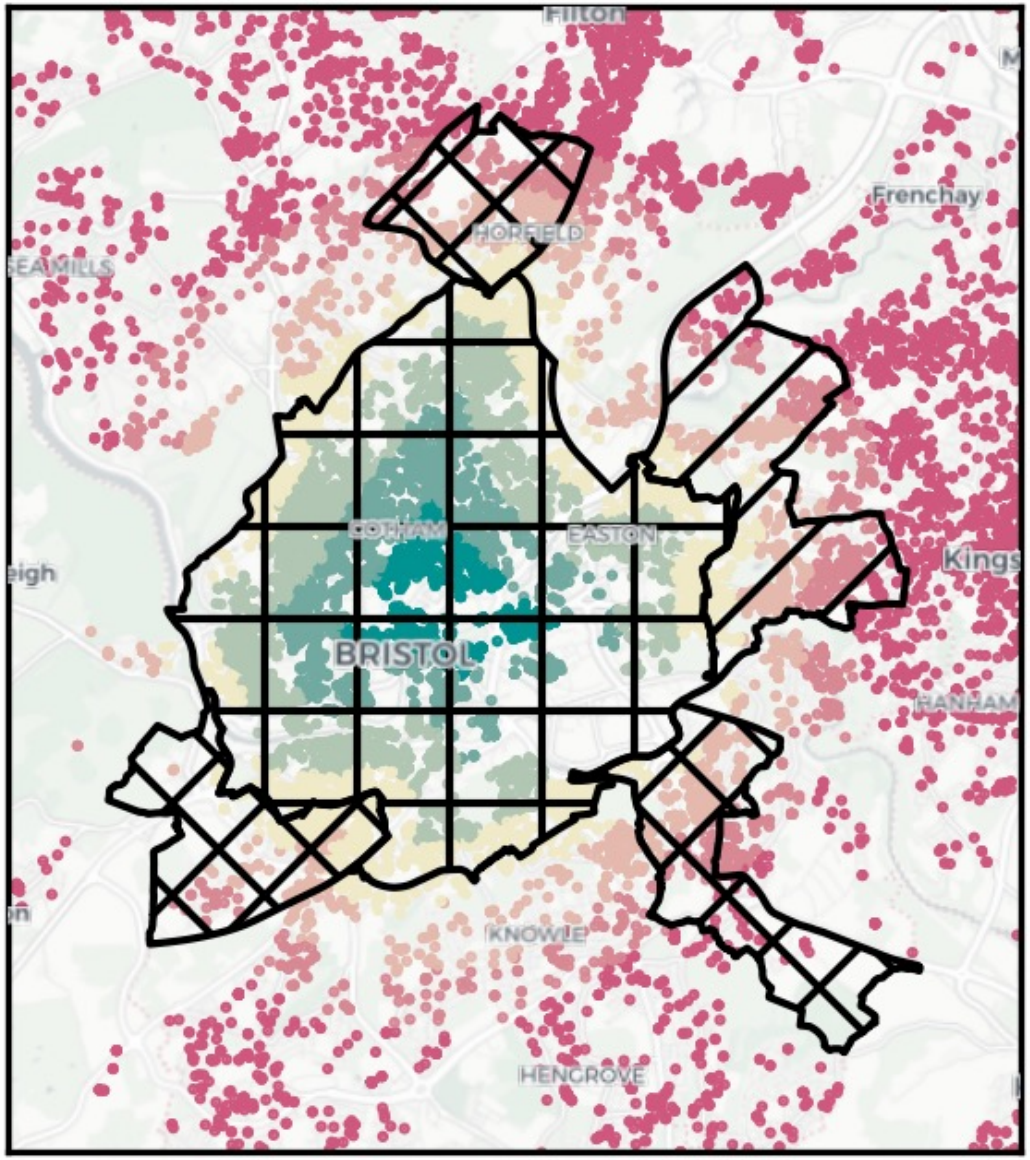
Listings near DLAs



Given the geography of these regulation areas, how can we be sure that their imposition is reflected in the “total” city rates?

Displacements **within** cities matter!

Listings near DLAs



Given the geography of these regulation areas, how can we be sure that their imposition is reflected in the "total" city rates?

Displacements **within** cities matter!

Neighborhood-level rent regulation distorts *local*, not city-wide, incentives.

We want to know: for similar properties, did licensing make a difference *on the margin?*

That is, **at the boundary of the DLA,** can we detect the difference the DLA makes?

WHAT & WHY ARE DLA?

IDENTIFICATION & ESTIMATION

PRELIMINARY RESULTS

	Pre-Treatment	Post Treatment
Inside	Pretreated	Treated
Outside	Pre-substitute	Substitute

IDENTIFICATION STRATEGIES

	Pre-Treatment	Post Treatment
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Treated listings inside DLA after treatment.
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We know city center (**T**,**PT**) differs from city edge (**S**,**PS**).

IDENTIFICATION STRATEGY 1: SDIDc

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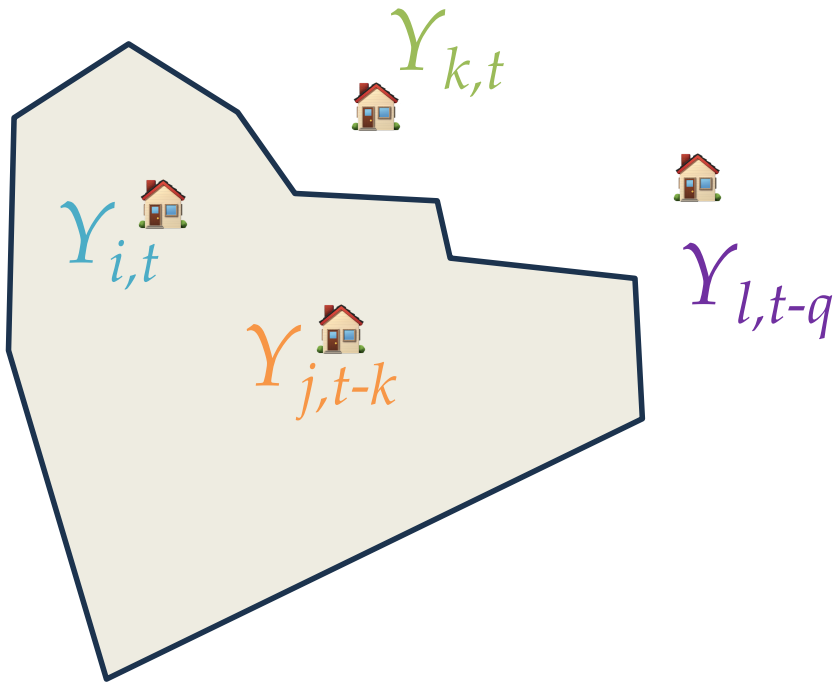
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(**T** – **S**) - (**PT** – **PS**) is the causal effect!

IDENTIFICATION STRATEGY 1: SDIDc



Borrow counterfactual from *just over* the boundary

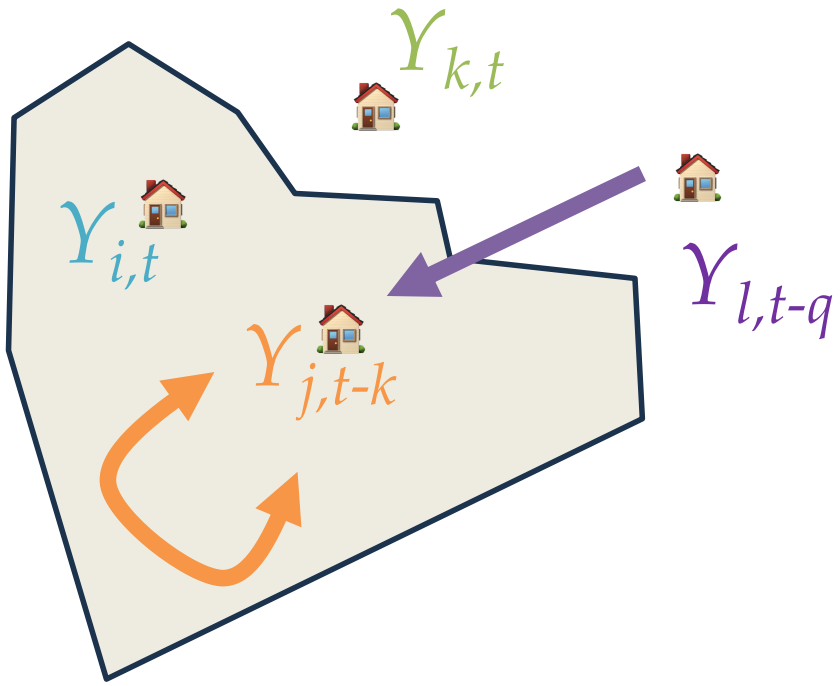
Pretreatment model:
fit on Y_j POs from Y_l

Treatment model:
fit on Y_i POs from Y_k

Substitute model:
fit on Y_k POs from Y_i

Pre-substitute model:
fit on Y_l POs from Y_j

IDENTIFICATION STRATEGY 1: SDIDc



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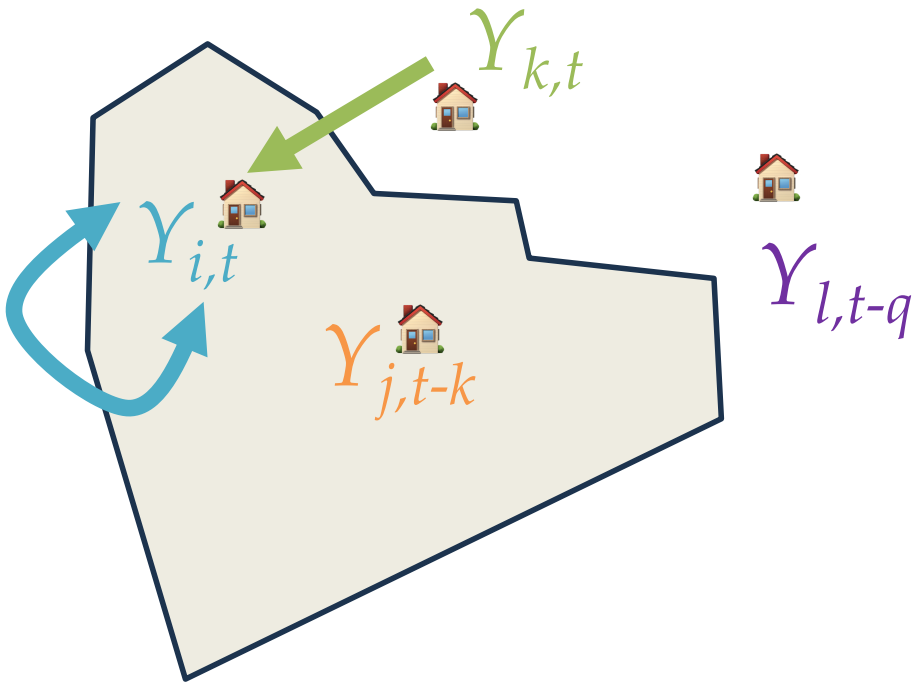
Treatment model:
fit on Y_i POs from Y_k

Substitute model:
fit on Y_k POs from Y_i

Pre-substitute model:
fit on Y_l POs from Y_j

Discontinuity: predicted outcome under "pre-treated" inside model vs. "never-treated" outside model

IDENTIFICATION STRATEGY 1: SDIDc



Pretreatment model:
fit on Y_j POs from Y_l

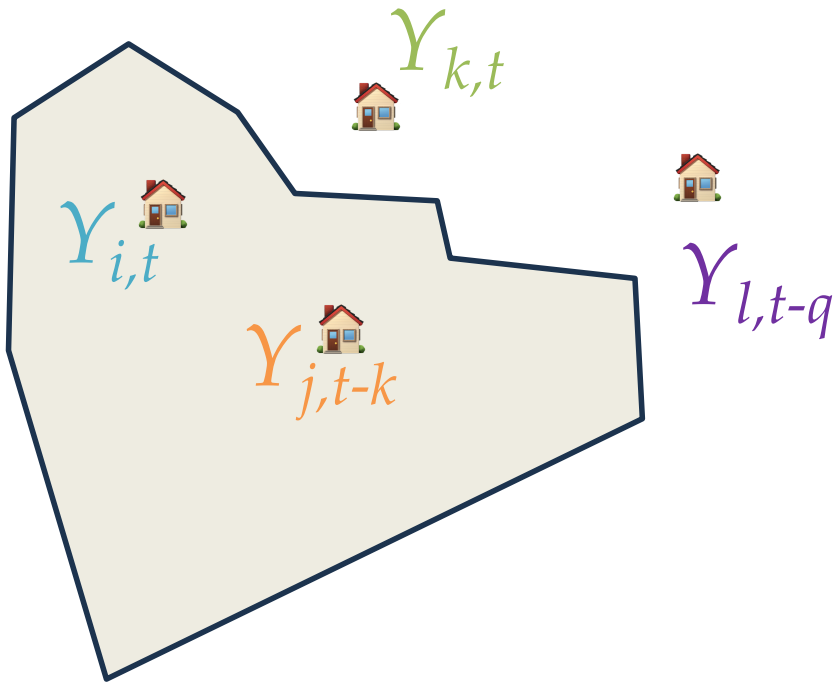
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Discontinuity: predicted outcome under "treated" inside model vs. "never-treated" outside model

IDENTIFICATION STRATEGY 1: SDIDc



Pretreatment model:
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ATE:
(**T** – **S**) - (**PT** – **PS**)

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(**T** – **S**) - (**PT** – **PS**) is the causal effect!

Estimated discontinuity posttreatment

Estimated discontinuity pretreatment

IDENTIFICATION STRATEGY: DOUBTS

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Estimated discontinuity posttreatment

Estimated discontinuity pretreatment

THIS IS JUST A COMPARISON ACROSS GROUP AVERAGES. WITHIN PROPERTY CHANGE IS MORE ROBUST.

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(**T** – **S**) - (**PT** – **PS**) is the causal effect!
T – **S** – **PT** + **PS**
T – **PT** – **S** + **PS**
(**T** – **PT**) - (**S** – **PS**)

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$$(T - S) - (PT - PS)$$
$$T - S - PT + PS$$
$$T - PT - S + PS$$
$$(T - PT) - (S - PS)$$

Estimated change inside — Estimated change outside

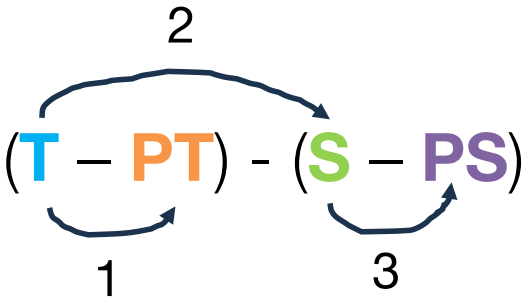
THIS IS JUST A COMPARISON ACROSS GROUP AVERAGES. WITHIN PROPERTY CHANGE IS MORE ROBUST. LETS DO IT!

IDENTIFICATION STRATEGY 2: QUARTET + X

Treated listings inside DLA after treatment.
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Pre-Substitute listings just outside DLA before treatment.

Matches needed:

1. treated to predecessor—match back in time w/in DLA
2. treated to substitute—match contemporary to out of DLA
3. substitute to predecessor—match back in time out of DLA

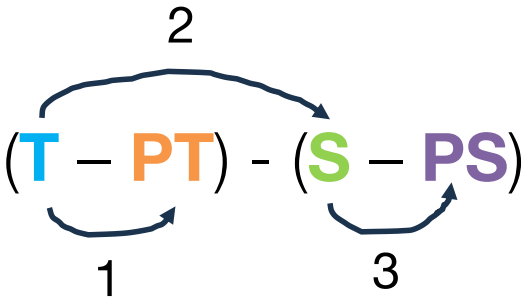


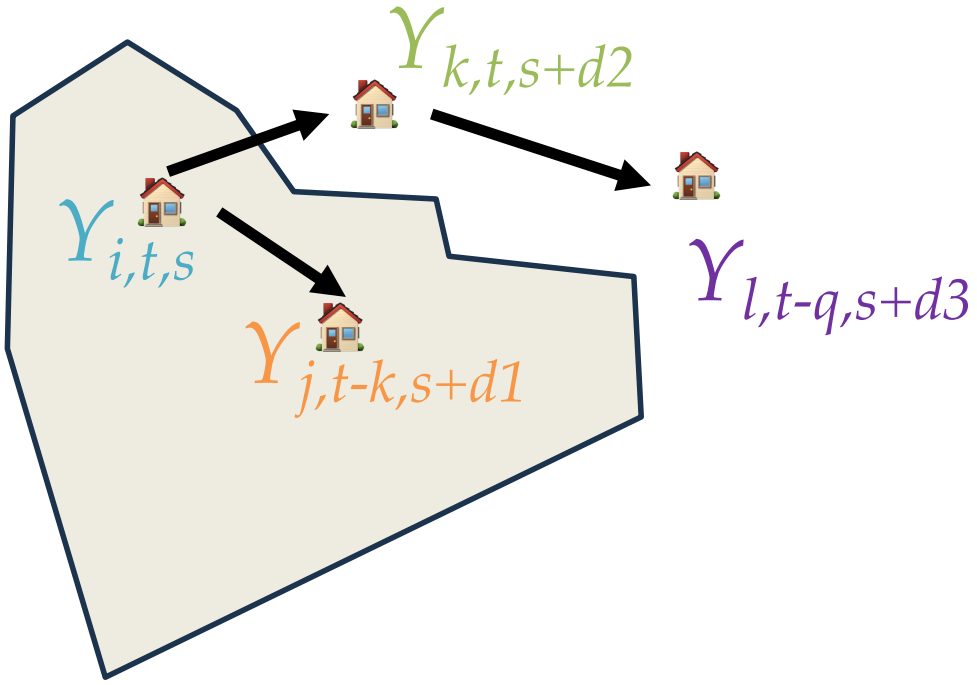
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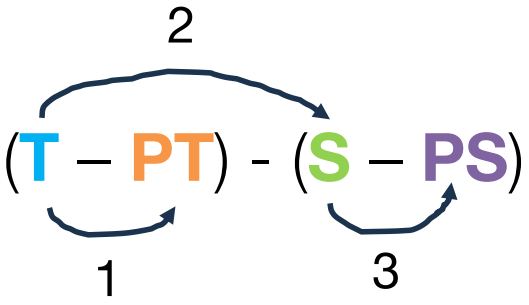


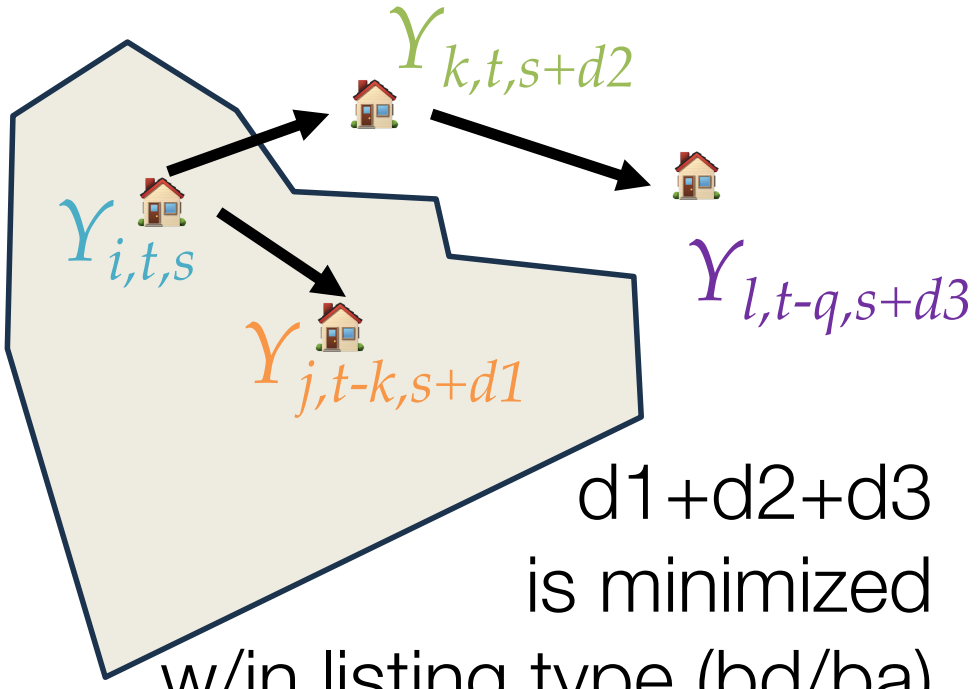
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d1+d2+d3
is minimized
w/in listing type (bd/ba)

IDENTIFICATION STRATEGY 2: QUARTET + X

(1) T→PT (2) T→S (3) S→PS

$$\textit{minimize} \quad \Sigma\, d_{ij} t_{ij} + \Sigma\, d_{ik} s_{ik} + \Sigma\, d_{kl} p_{kl}$$

subject to

$$\Sigma_j\, t_{ij} = 1$$ *all treatments get one pretreatment*

$$\Sigma_i\, t_{ij} \leq 1$$ *use pretreatments at most once*

$$\Sigma_k\, s_{ik} = 1$$ *all treatments get one substitute*

$$\Sigma_l\, p_{kl} \geq \Sigma_j\, s_{jk}$$ *every selected substitute gets a presub*

$$\Sigma_i\, s_{ik} \leq 1$$ *use substitutes at most once*

$$\Sigma_k\, p_{kl} \leq 1$$ *use presubs at most once*

IDENTIFICATION STRATEGY 2: QUARTET + X

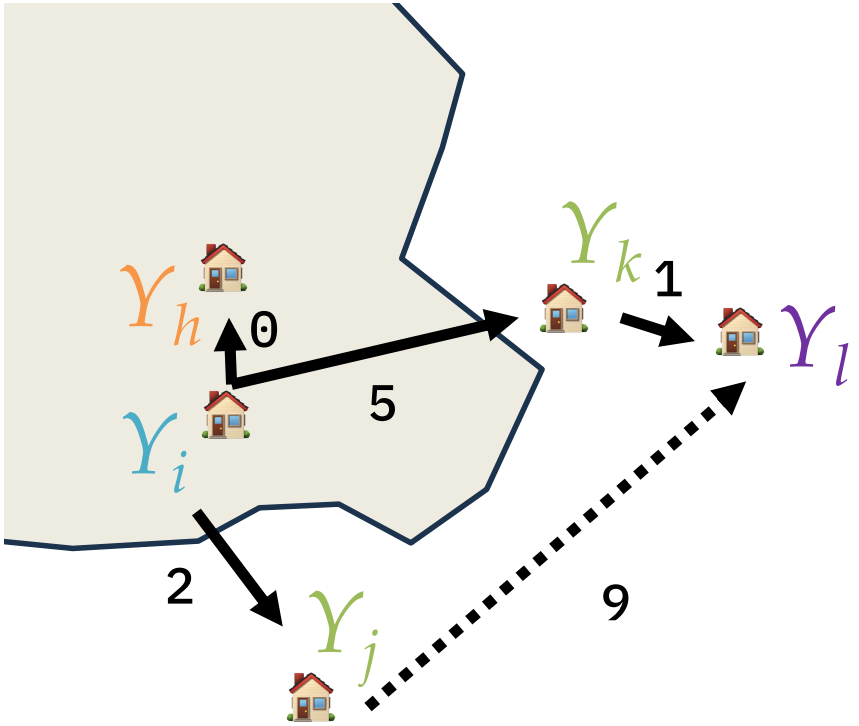
(1) T→PT (2) T→S (3) S→PS

minimize $\sum d_{ij}t_{ij} + \sum d_{ik}s_{ik} + \sum d_{kl}p_{kl}$

subject to

Inside subproblem	$\sum_j t_{ij} = 1$	*all treatments get one pretreatment*
	$\sum_i t_{ij} \leq 1$	*use pretreatments at most once*
Outside subproblem	$\sum_k s_{ik} = 1$	*all treatments get one substitute*
	$\sum_l p_{kl} \geq \sum_j s_{jk}$	*every selected substitute gets a presub*
	$\sum_i s_{ik} \leq 1$	*use substitutes at most once*
	$\sum_k p_{kl} \leq 1$	*use presubs at most once*

IDENTIFICATION STRATEGY 2: QUARTET + X



Greedy optimization picks <hijl> for a cost of 11

Joint optimization picks <hikl> for a cost of 6!

Indeed, j might go unused w/o a nearer presub!

IDENTIFICATION STRATEGY 2: QUARTET + X

$$\Upsilon_i - \Upsilon_h = \delta_i = f(X_i) + e_i$$ TREATED AREA MODEL

$$\Upsilon_k - \Upsilon_l = \delta_k = f(X_k) + e_k$$ UNTREATED AREA MODEL

$$\tau_1 = \delta_1 - \hat{\delta}_0 = f(X) + u$$ TREATED PO MODEL

$$\tau_0 = \hat{\delta}_1 - \delta_0 = f(X) + u$$ UNTREATED PO MODEL

$$p(t = 1) = f(X) + v$$ PROPENSITY MODEL

ATE: $n^{-1} \Sigma \, \tau_0 p + \tau_1 (1\text{-}p)$

IDENTIFICATION STRATEGY 2: QUARTET + X

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Predict the change in price for properties inside the DLA/outside of the DLA.

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$$\tau_0 = \hat{\delta}_1 - \delta_0 = f(X) + u$$ UNTREATED PO MODEL

δ_1: *Observed inside change*
$\hat{\delta}_0$: *Outside's prediction about an inside change*
δ_0: *Observed outside change*
$\hat{\delta}_1$: *Inside's prediction about an outside change*

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Predict the difference between observed change & counterfactual predicted change

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$$p(t = 1) = f(X) + v$$ PROPENSITY MODEL

How likely is a property with X treated?

IDENTIFICATION STRATEGY 2: QUARTET + X

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$$p(t = 1) = f(X) + v$$ PROPENSITY MODEL

Mean PO gap, weighted by treatment likelihood

ATE: $n^{-1}\Sigma\ \tau_0 p + \tau_1 (1\text{-}p)$

IDENTIFICATION STRATEGY 2: QUARTET + X

WHAT & WHY ARE DLA?

IDENTIFICATION & ESTIMATION

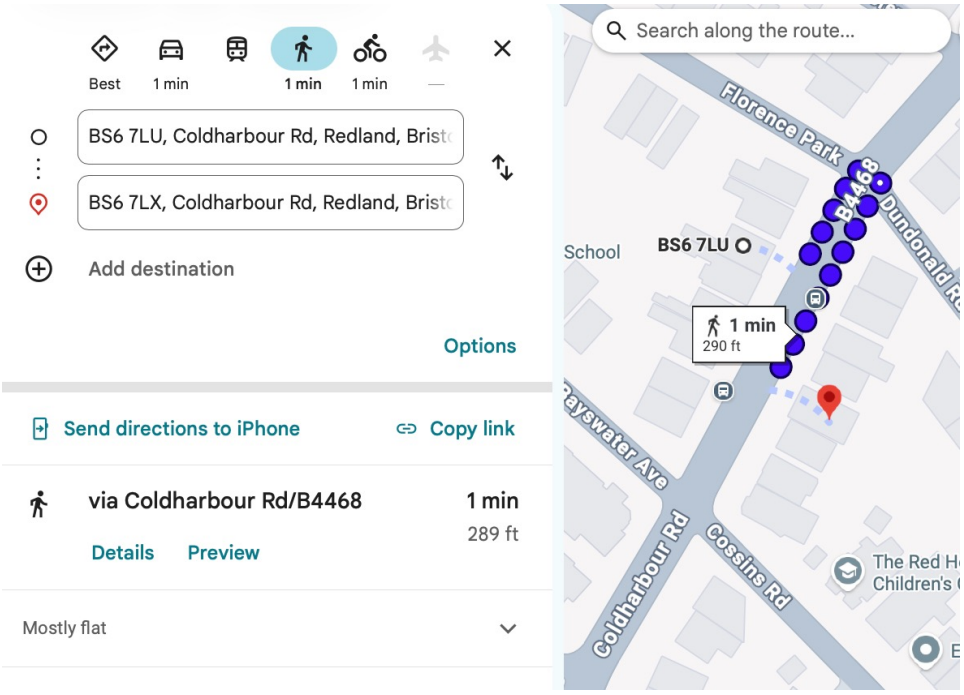
PRELIMINARY RESULTS

SLA OR NAY?

kind	uid	Treated	In DLA?	Date	Postcode	Beds	Baths	Rent
treated	7459f71781	TRUE	TRUE	2019-08-24	BS67LX	1	1	167
pretreat	777d5a2831	FALSE	TRUE	2018-07-28	BS67LX	1	1	162
substitute	ea79ced515	FALSE	FALSE	2019-06-06	BS67LU	1	1	150
presubstitute	cd861f2032	FALSE	FALSE	2018-06-29	BS67LU	1	1	144

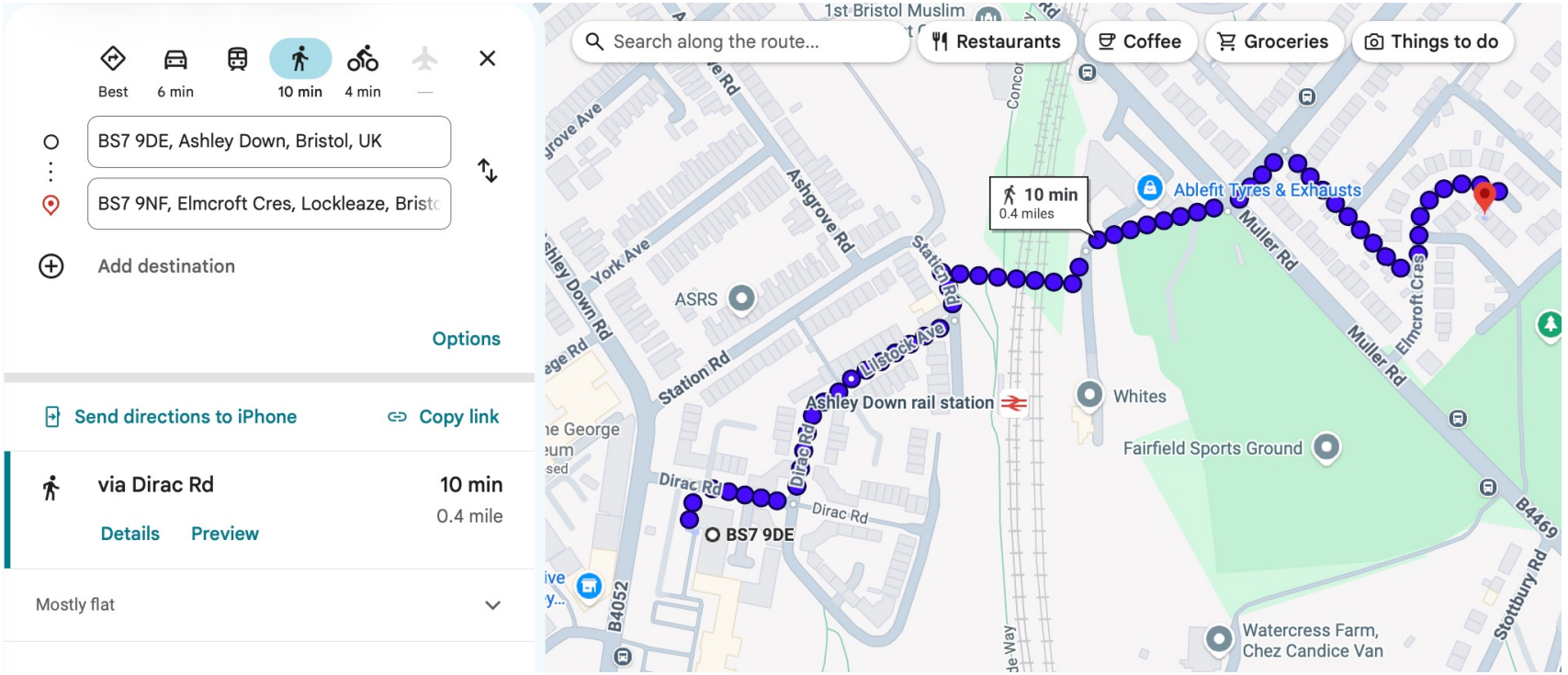
MATCH EXAMPLE

kind	uid	Treated	In DLA?	Date	Postcode	Beds	Baths	Rent
treated	7459f71781	TRUE	TRUE	2019-08-24	BS67LX	1	1	167
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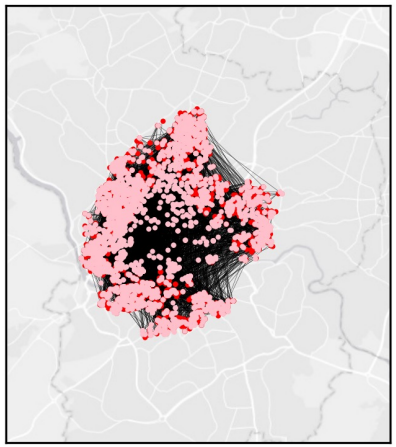
MATCH EXAMPLE

kind	uid	Treated	In DLA?	Date	Postcode	Beds	Baths	Rent
treated	213afae5d7	TRUE	TRUE	2020-06-19	BS79DE	2	1	254
pretreat	2e11c99767	FALSE	TRUE	2019-04-25	BS79DE	2	1	231
substitute	6de3f08d32	FALSE	FALSE	2020-08-20	BS79NF	2	1	225
presubstitute	db60128988	FALSE	FALSE	2019-07-12	BS79NF	2	1	219

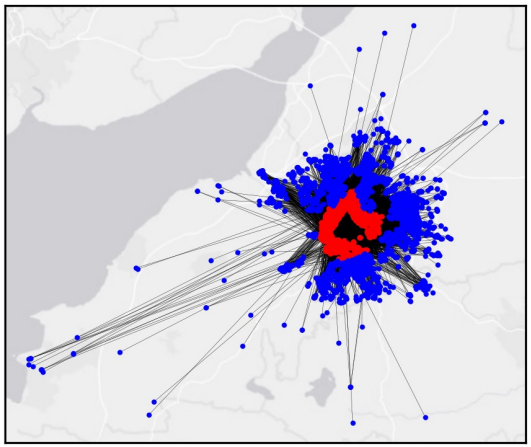


MATCH EXAMPLE

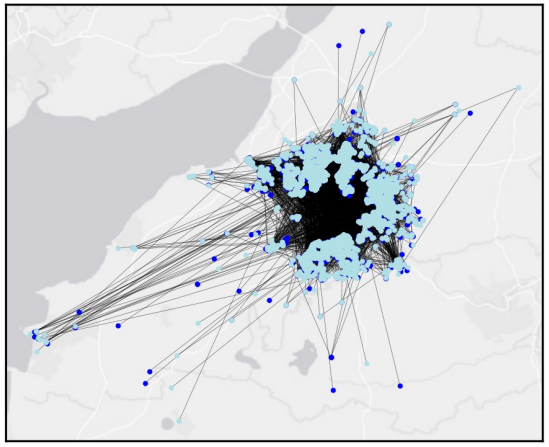
Treated to Predecessor

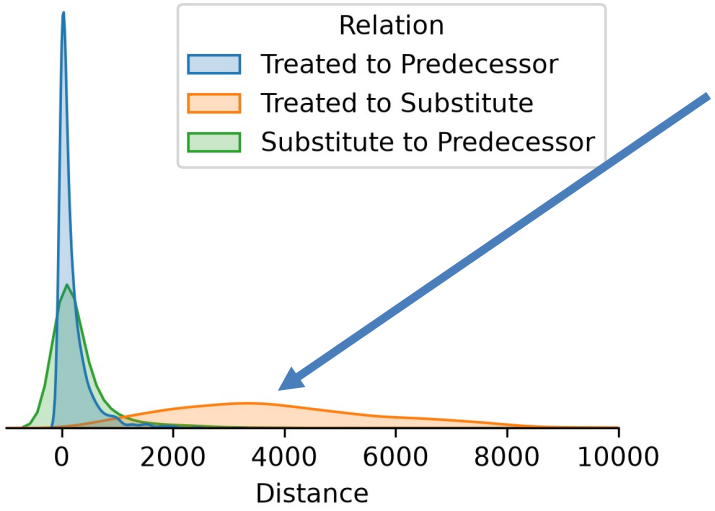


Treated to Substitute



Substitute to Predecessor



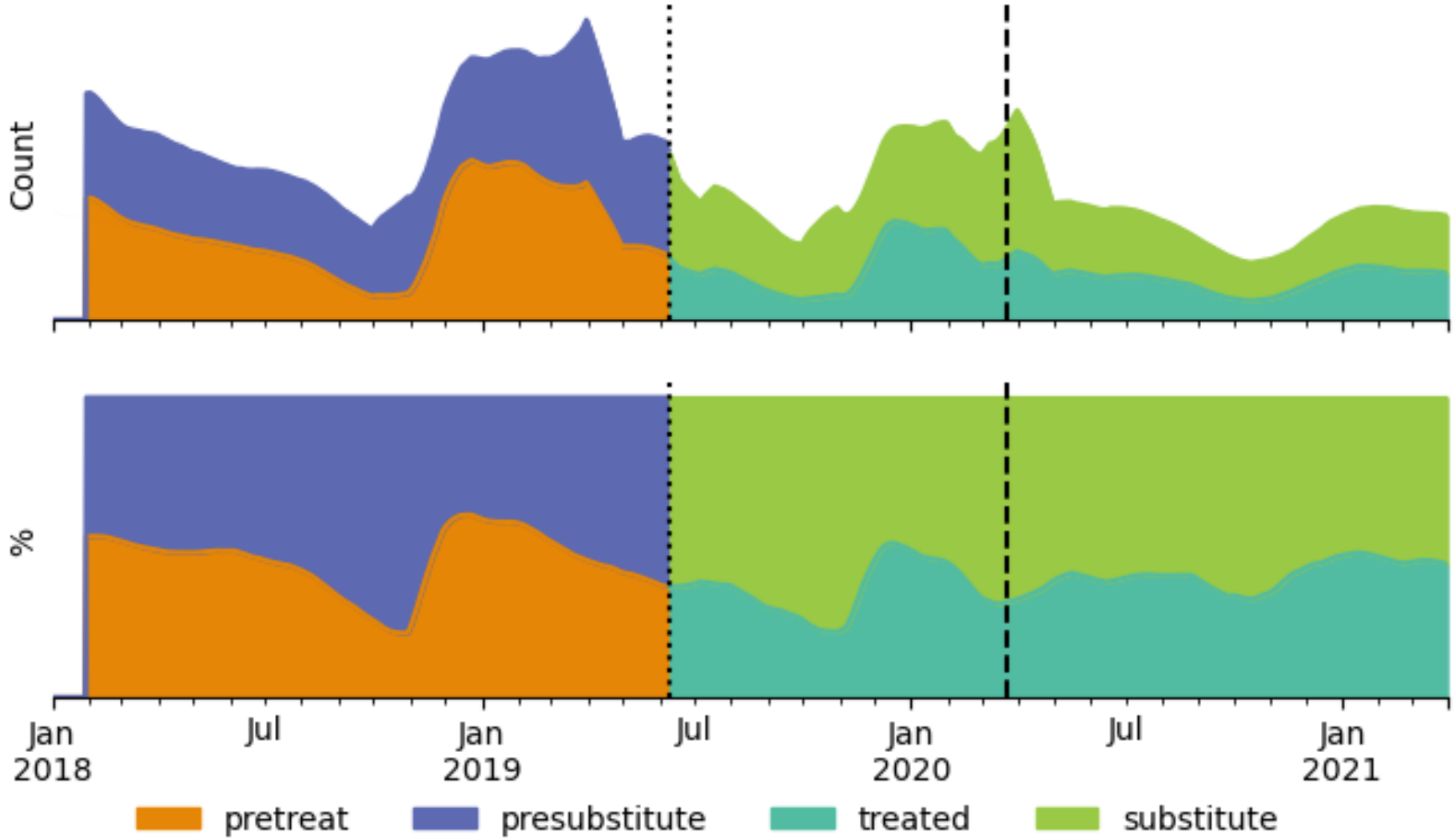


Proximity of the matching leaves something to be desired… but w/ ~5k matches, 729 are “local.”

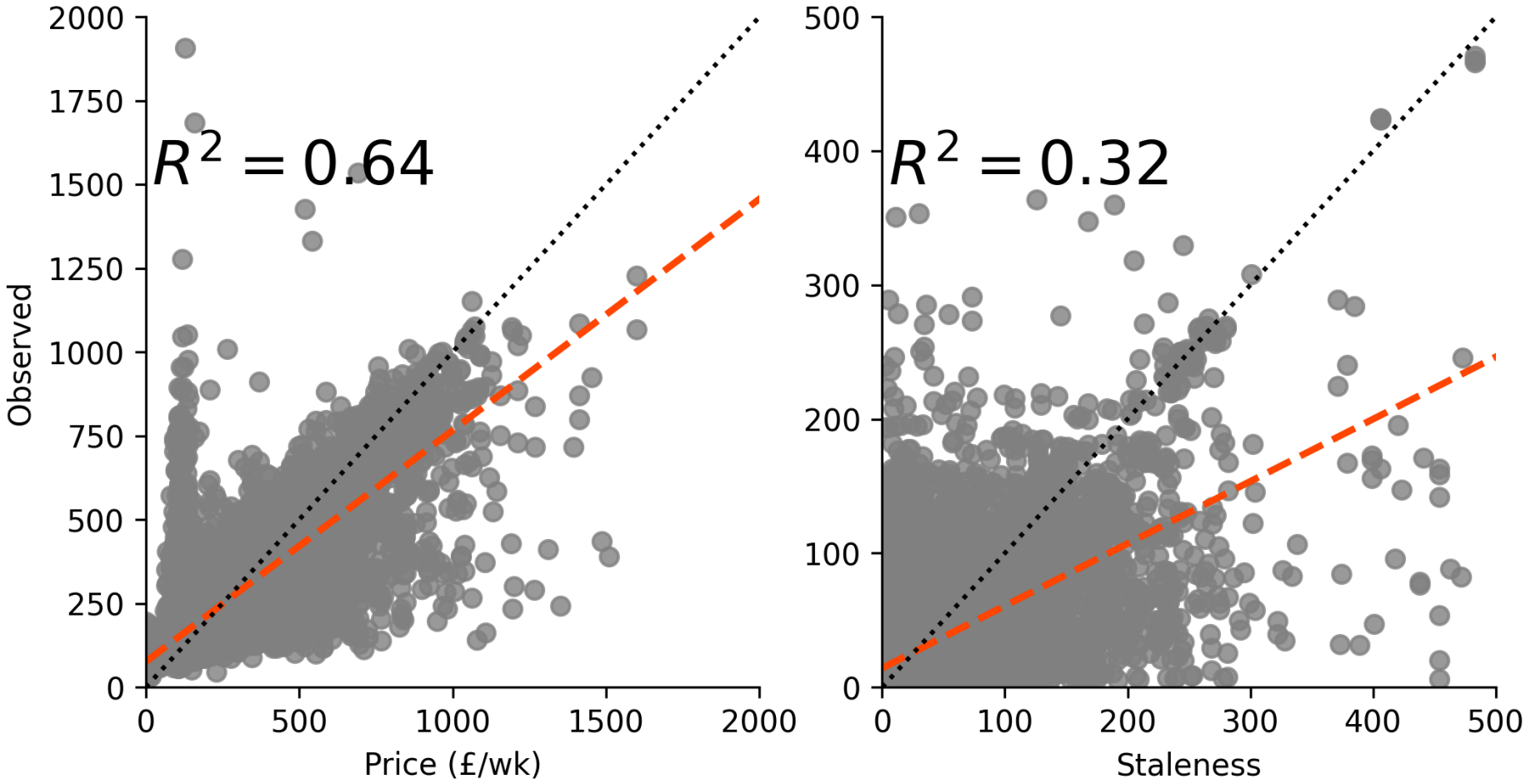
listings are 2018-2023
policy is 2019-2024

More data won’t solve the treated to substitute match

MATCH QUALITY



SUPPLY EFFECTS: TOTAL LISTINGS



SPATIAL DIFFERENCE IN DISCONTINUITY

	Overall		Boundary	
outcome	duration	price	duration	price
presubstitute	4.60	-37.03	-2.27	-17.30
pretreat	4.04	26.15	-0.38	20.49
substitute	2.51	-42.05	-6.33	-7.98
treated	4.48	22.07	-0.83	14.63

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Predicted price from a listing's own-side model minus the across-boundary model

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Models trained just inside the DLA boundary predict higher prices for the same houses…

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Models trained just inside the DLA boundary predict higher prices for the same houses...
gap narrows post-treatment.

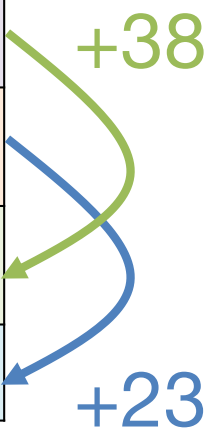
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	Overall		Boundary	
outcome	duration	price	duration	price
ATE	2.52	0.94	3.62	-15.17

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CHANGE IN THE GAP

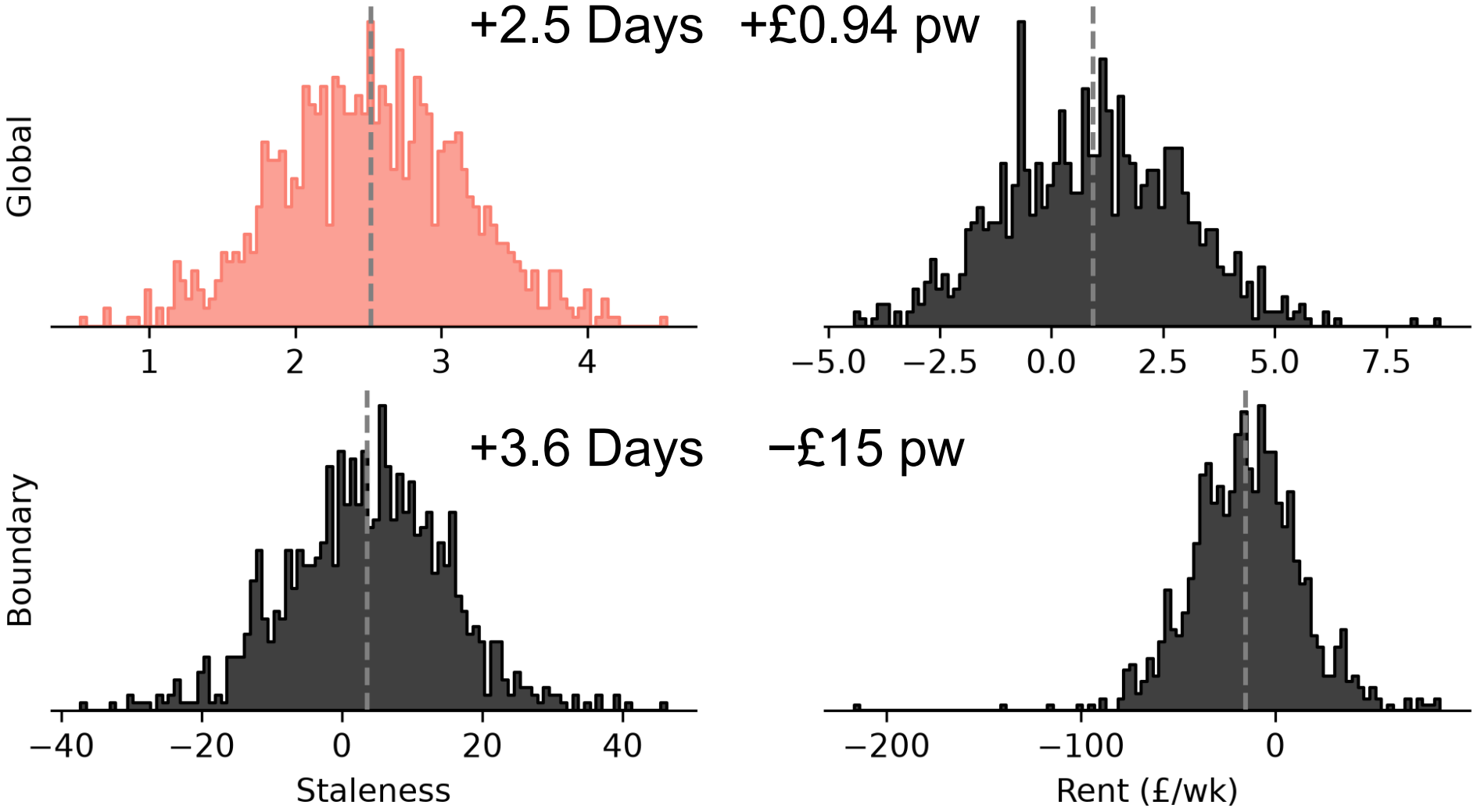
	Overall		Boundary	
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SPATIAL DIFFERENCE IN DISCONTINUITY

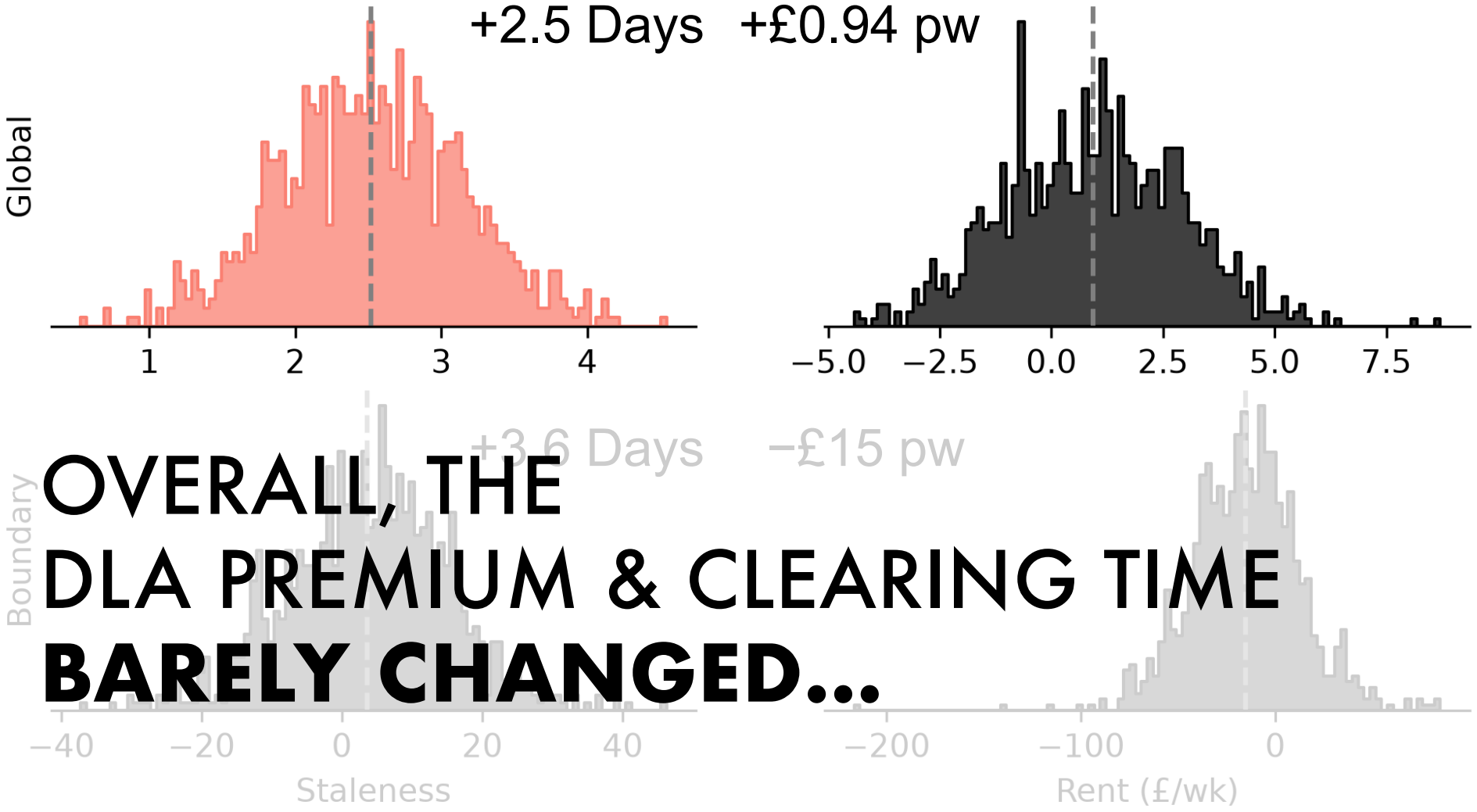
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outcome	duration	price	duration	price
ATE	2.52	0.94	3.62	-15.17

SDIDc: SMALLER PREMIUM AT THE BOUNDARY



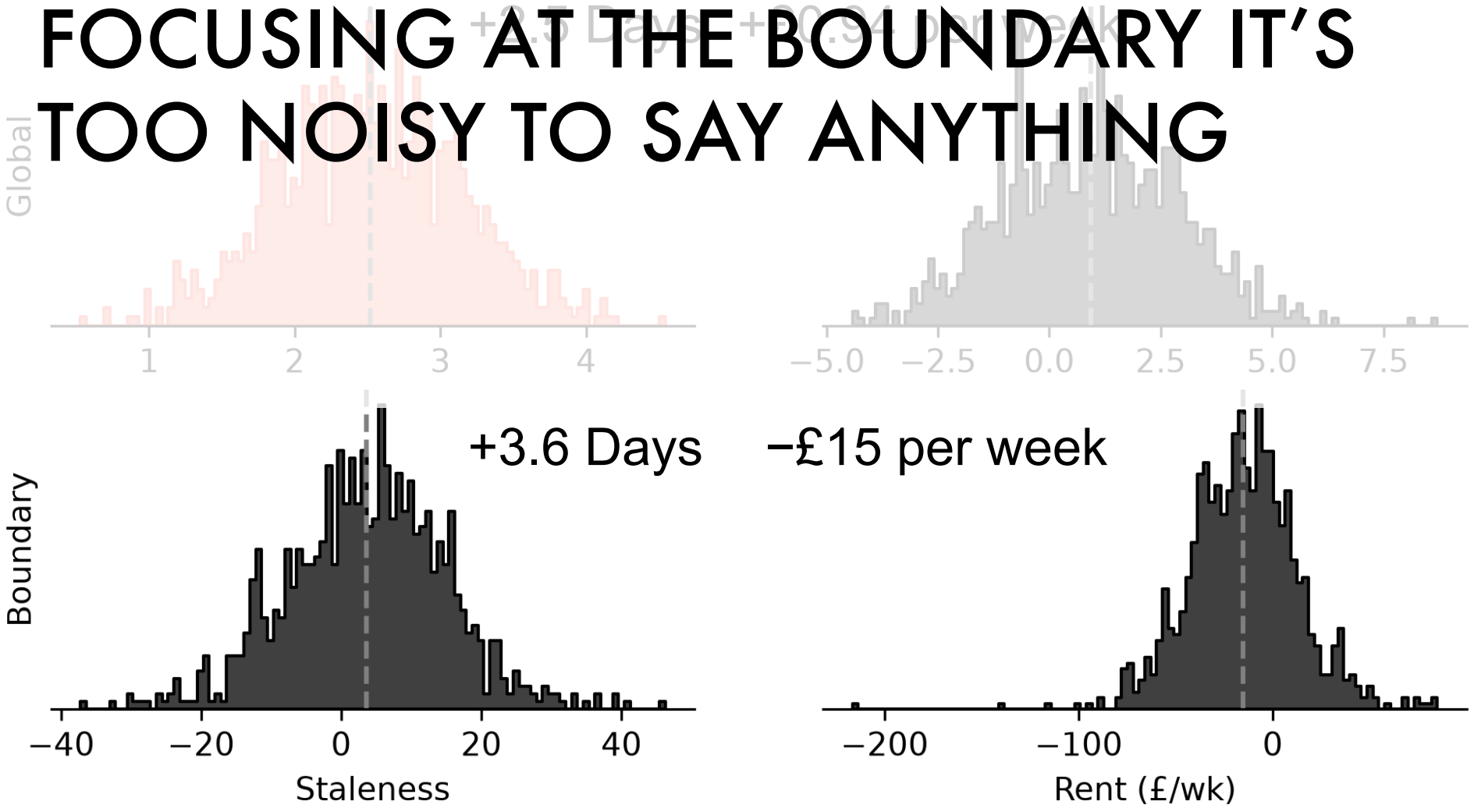
SPATIAL DIFFERENCE IN DISCONTINUITY



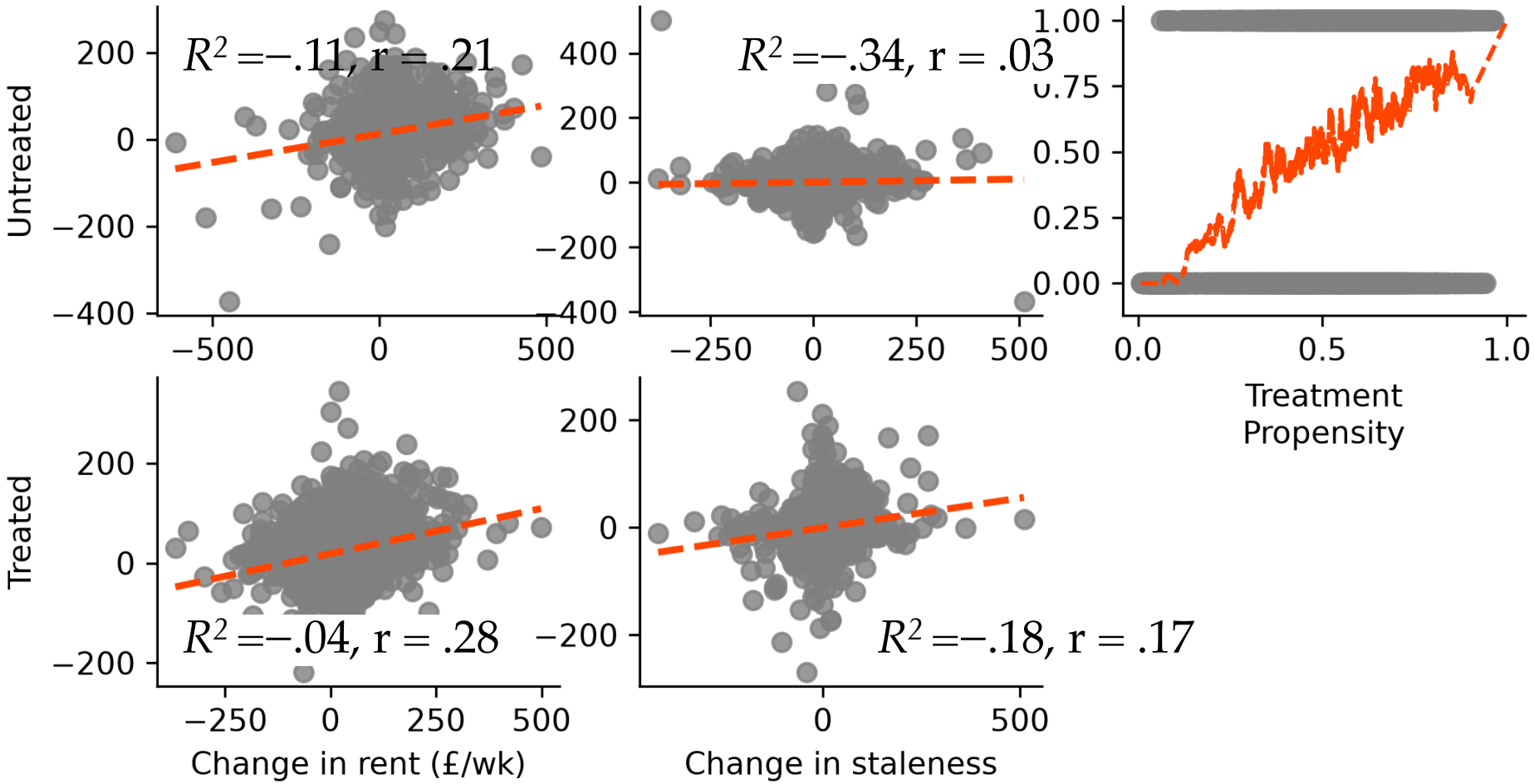
OVERALL, THE DLA PREMIUM & CLEARING TIME **BARELY CHANGED...**

SPATIAL DIFFERENCE IN DISCONTINUITY

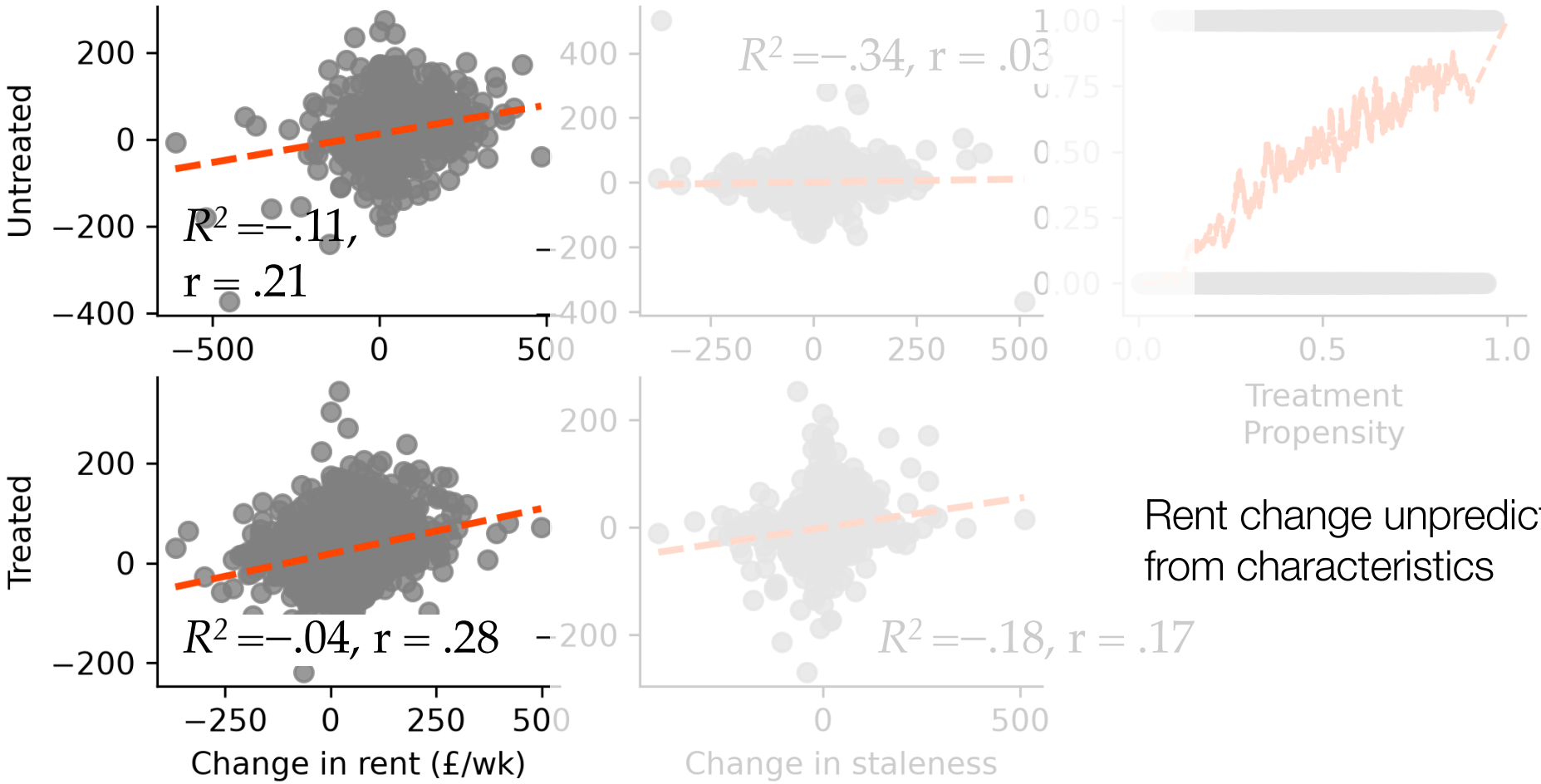
FOCUSING AT THE BOUNDARY IT'S TOO NOISY TO SAY ANYTHING



SPATIAL DIFFERENCE IN DISCONTINUITY

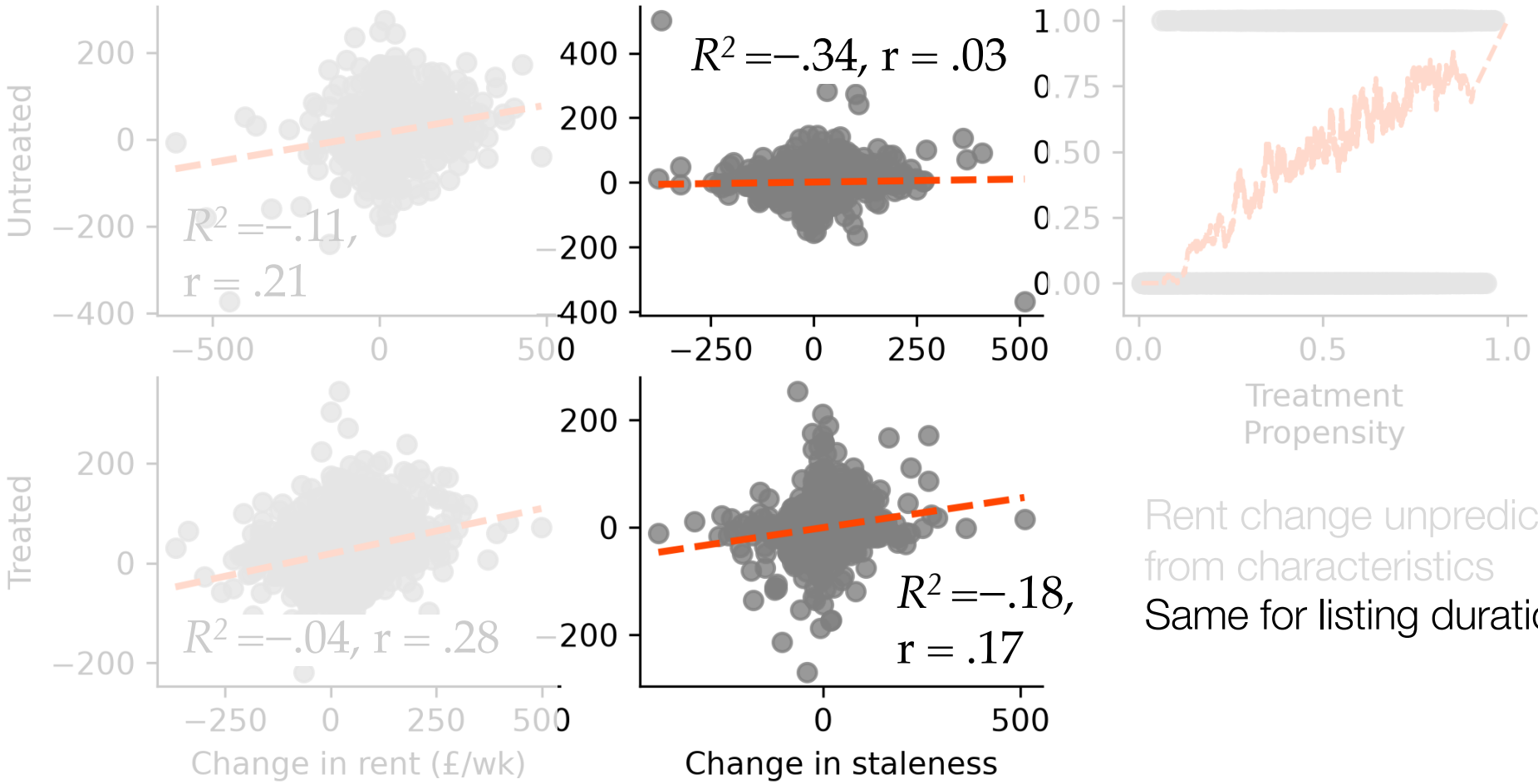

$R^2 = -.11$, r = .21
$R^2 = -.34$, r = .03
$R^2 = -.04$, r = .28
$R^2 = -.18$, r = .17
Untreated
Treated
Change in rent (£/wk)
Change in staleness
Treatment Propensity

X-LEARNER



Rent change unpredictable from characteristics

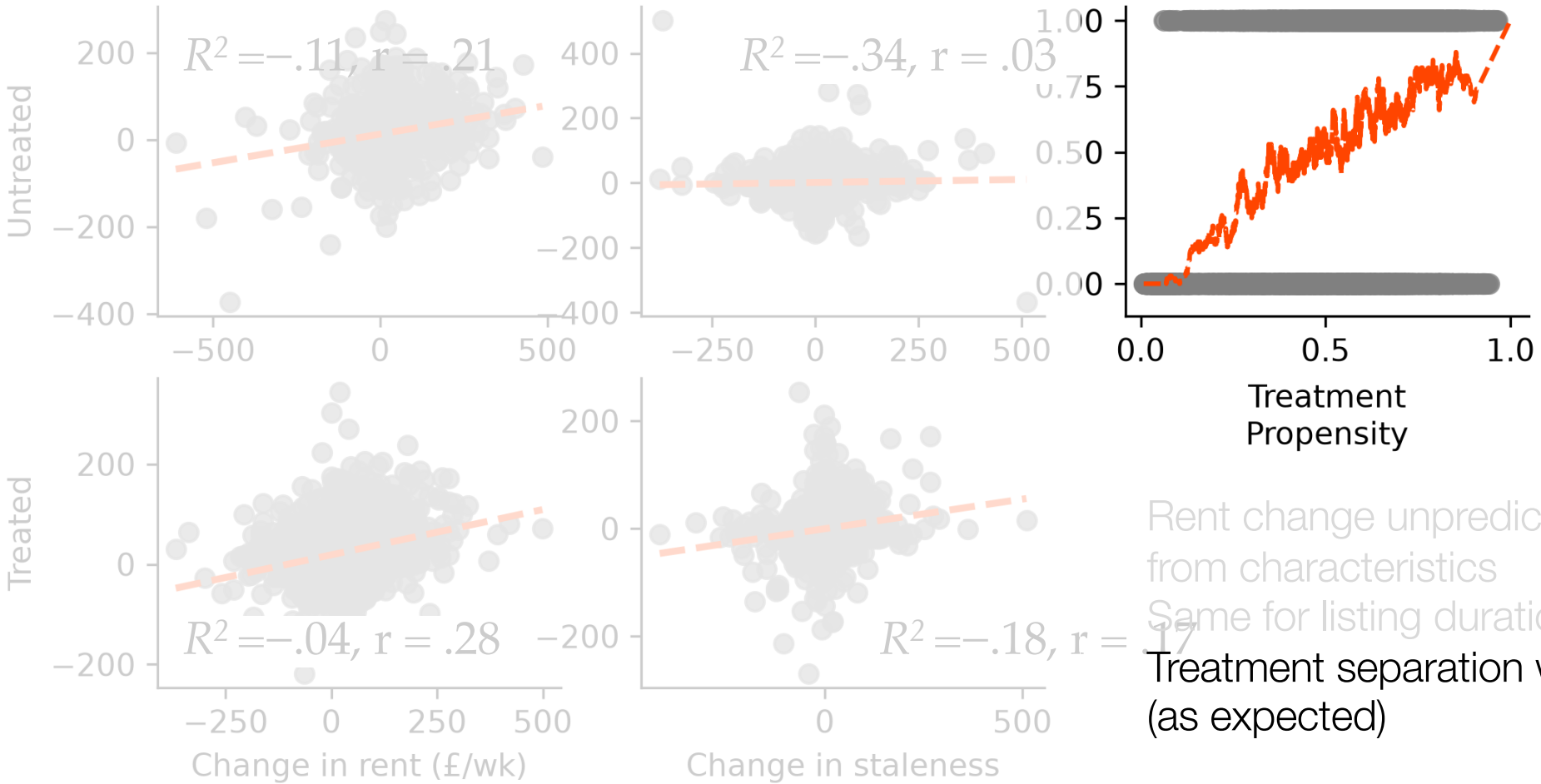
X-LEARNER



Rent change unpredictable from characteristics

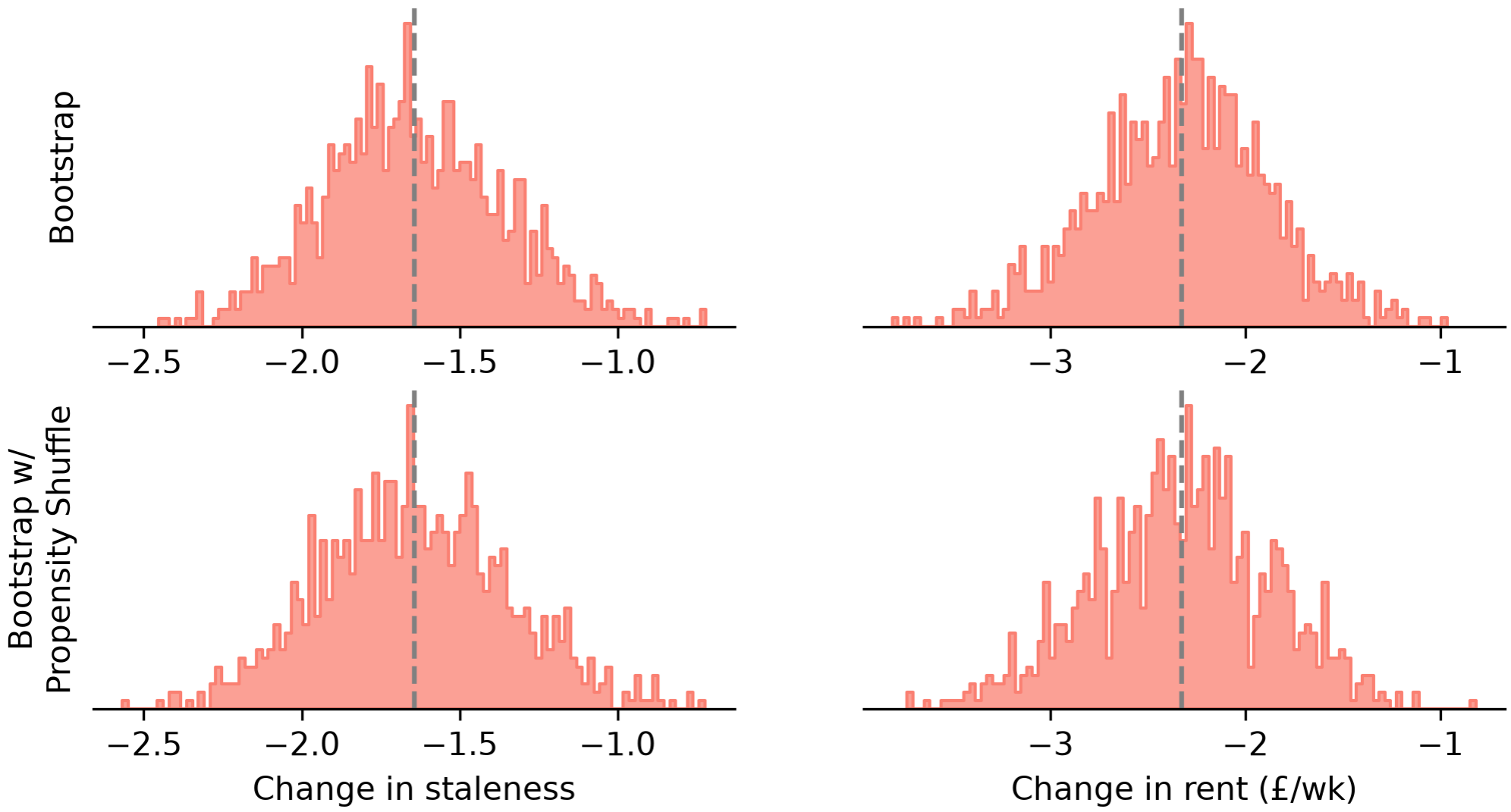
Same for listing duration…

X-LEARNER

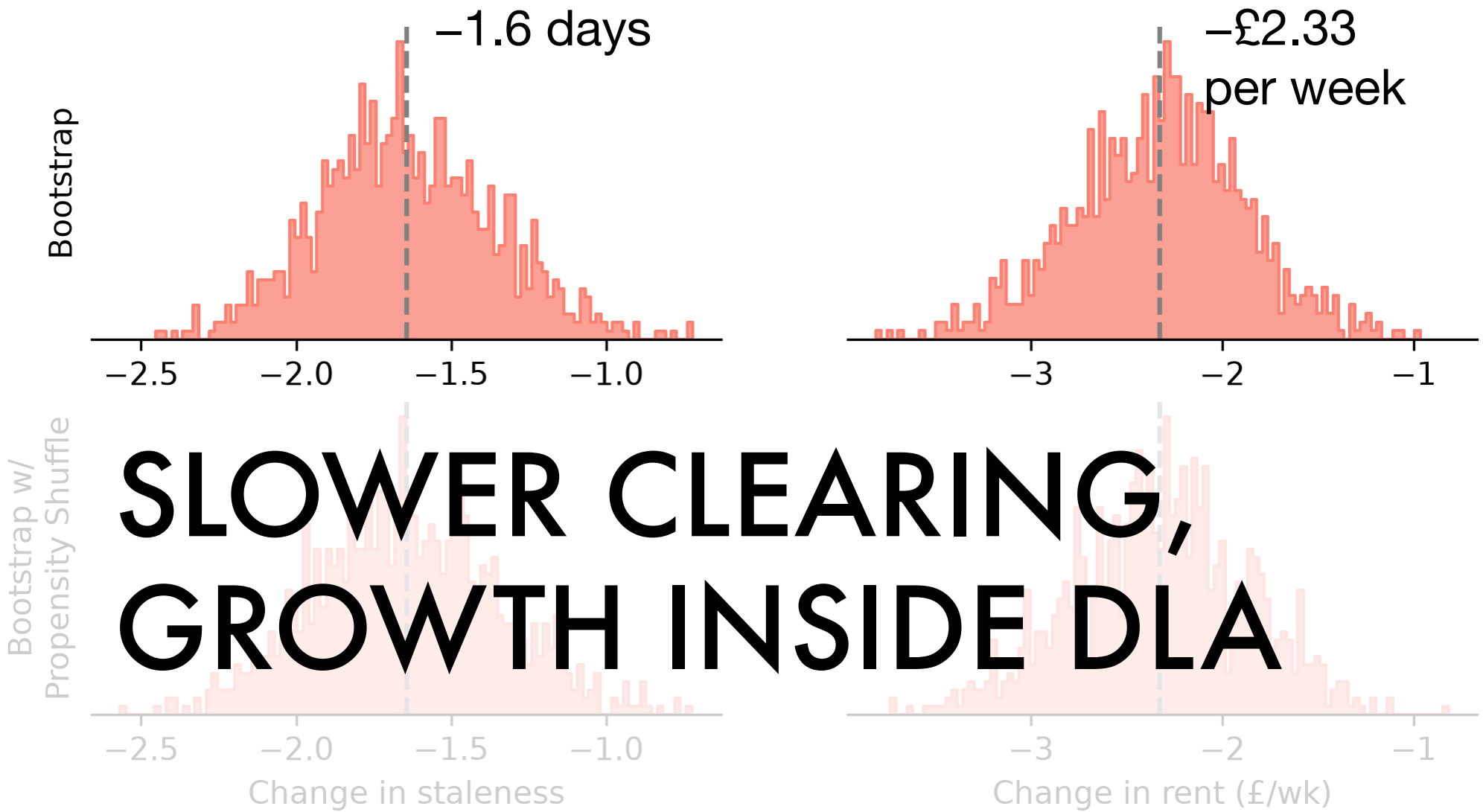


Rent change unpredictable from characteristics
Same for listing duration…
Treatment separation weak (as expected)

X-LEARNER


Bootstrap
Bootstrap w/
Propensity Shuffle
−2.5
−2.0
−1.5
−1.0
−3
−2
−1
Change in staleness
Change in rent (£/wk)

X-LEARNER



SLOWER CLEARING, GROWTH INSIDE DLA

X-LEARNER

(**T** – **S**) - (**PT** – **PS**) = (**T** – **PT**) - (**S** – **PS**) = ATE

SDIDc X-Learner

True in theory, but empirically…

rent: SDIDc noisy negative, X-Learner negative

clearing: SDIDc noisy positive, X-Leaner negative

Why?

SLA OR NAY?

(T – S) - (PT – PS) = (T – PT) - (S – PS) = ATE

SDIDc X-Learner

True in theory, but empirically…

rent: SDIDc noisy negative, X-Learner negative

clearing: SDIDc noisy positive, X-Leaner negative

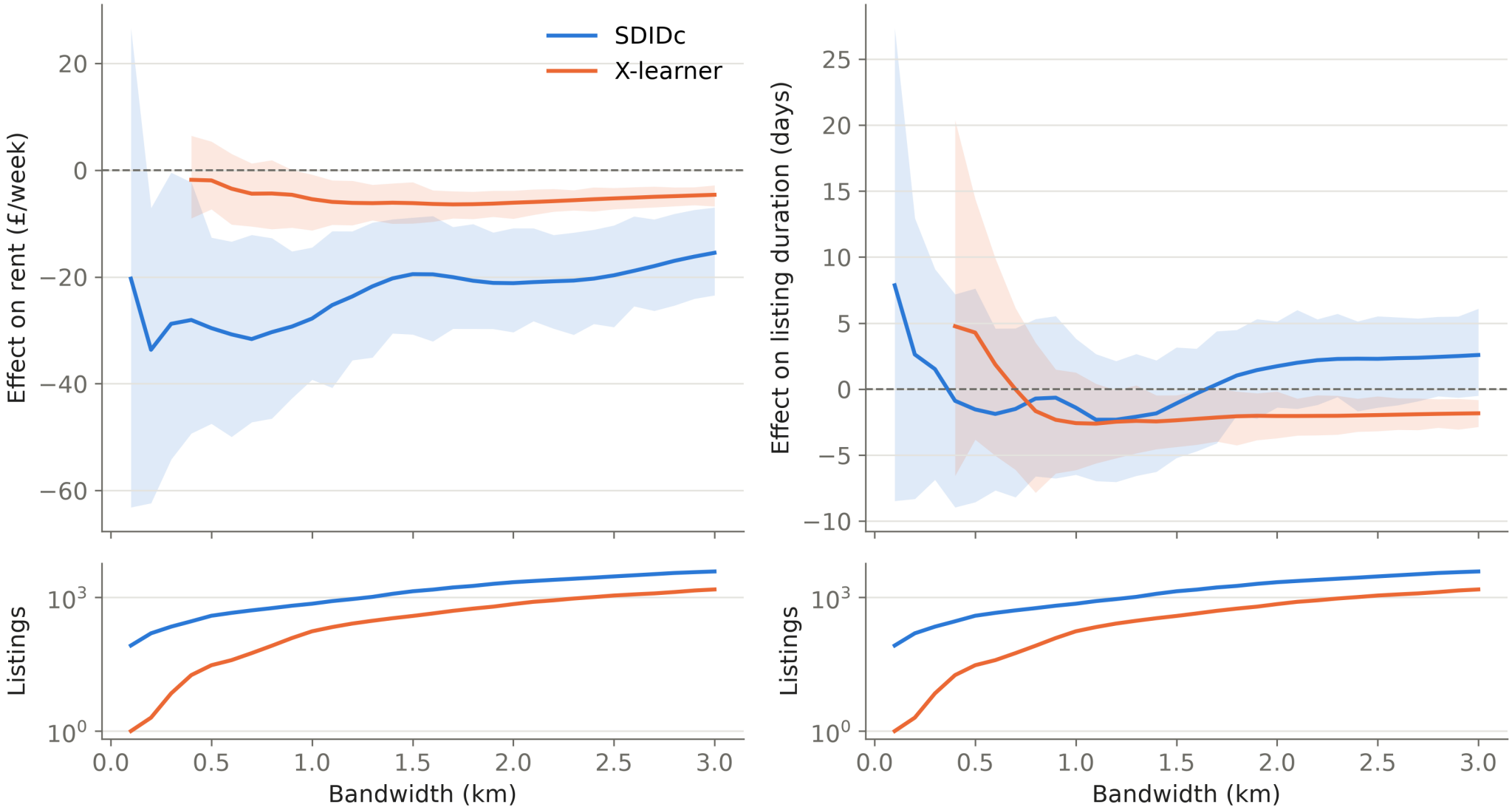
Why?

SDIDc: temporal difference of spatial differences

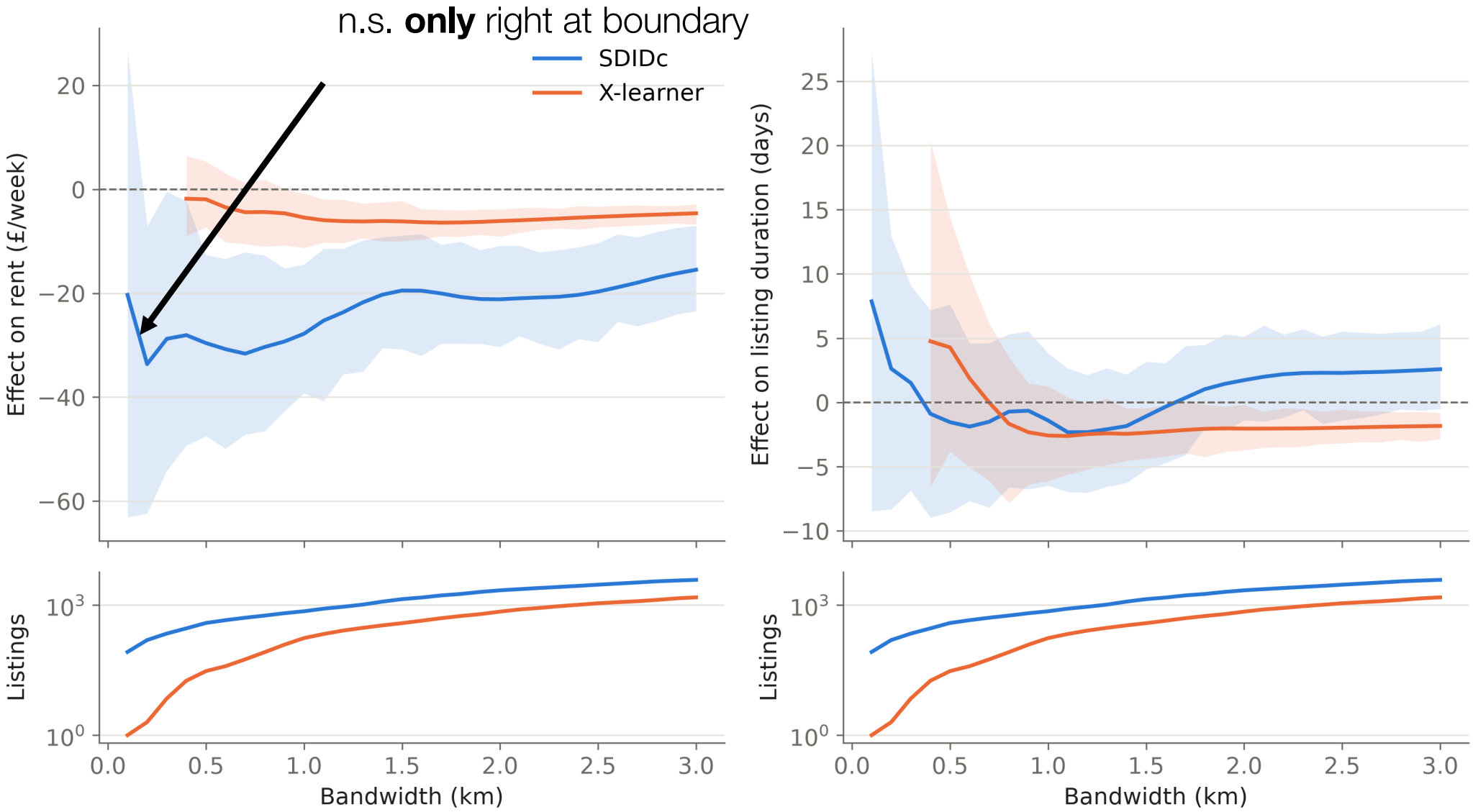
X-Learner: spatial difference of temporal differences

How space gets differenced matters!

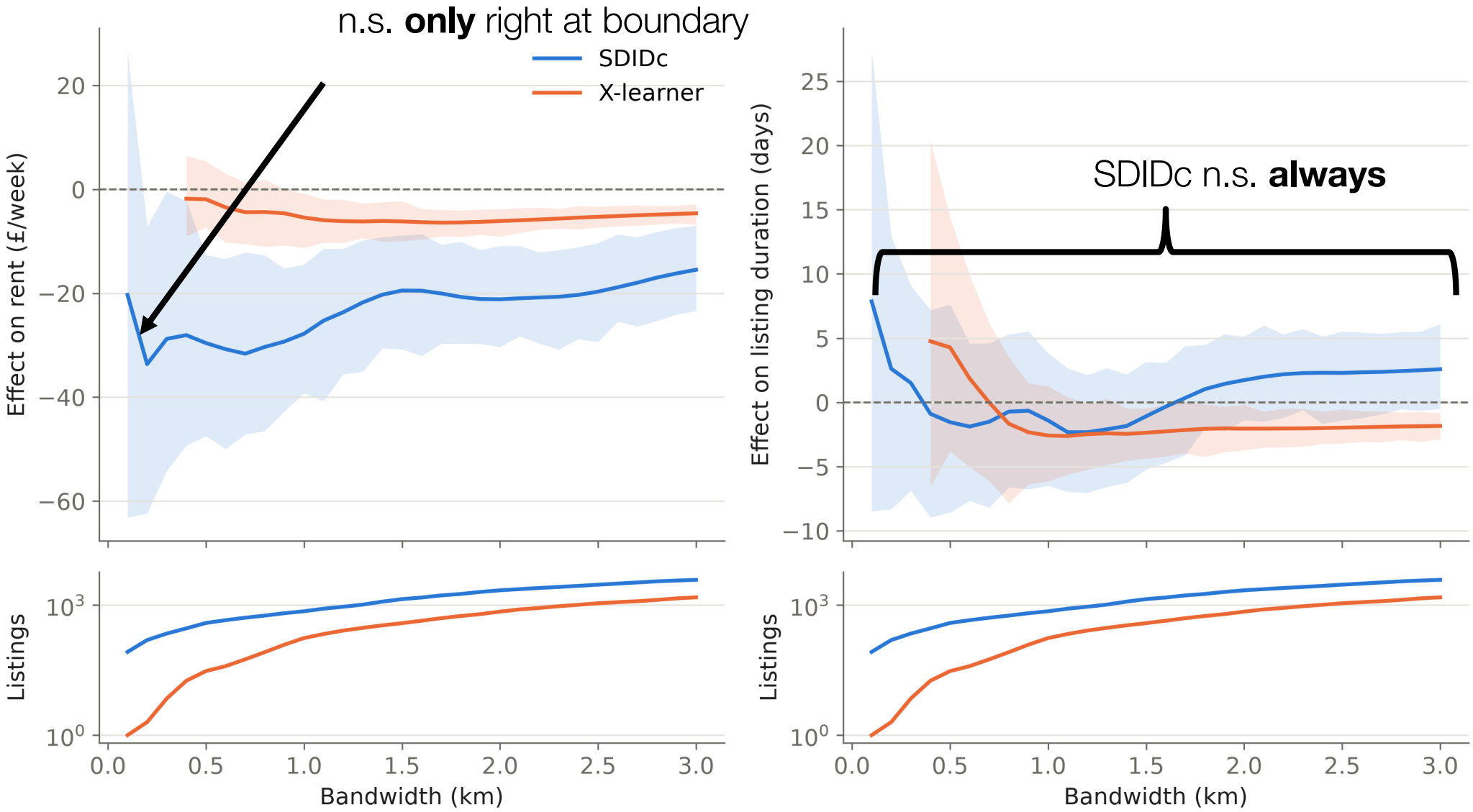
SLA OR NAY?



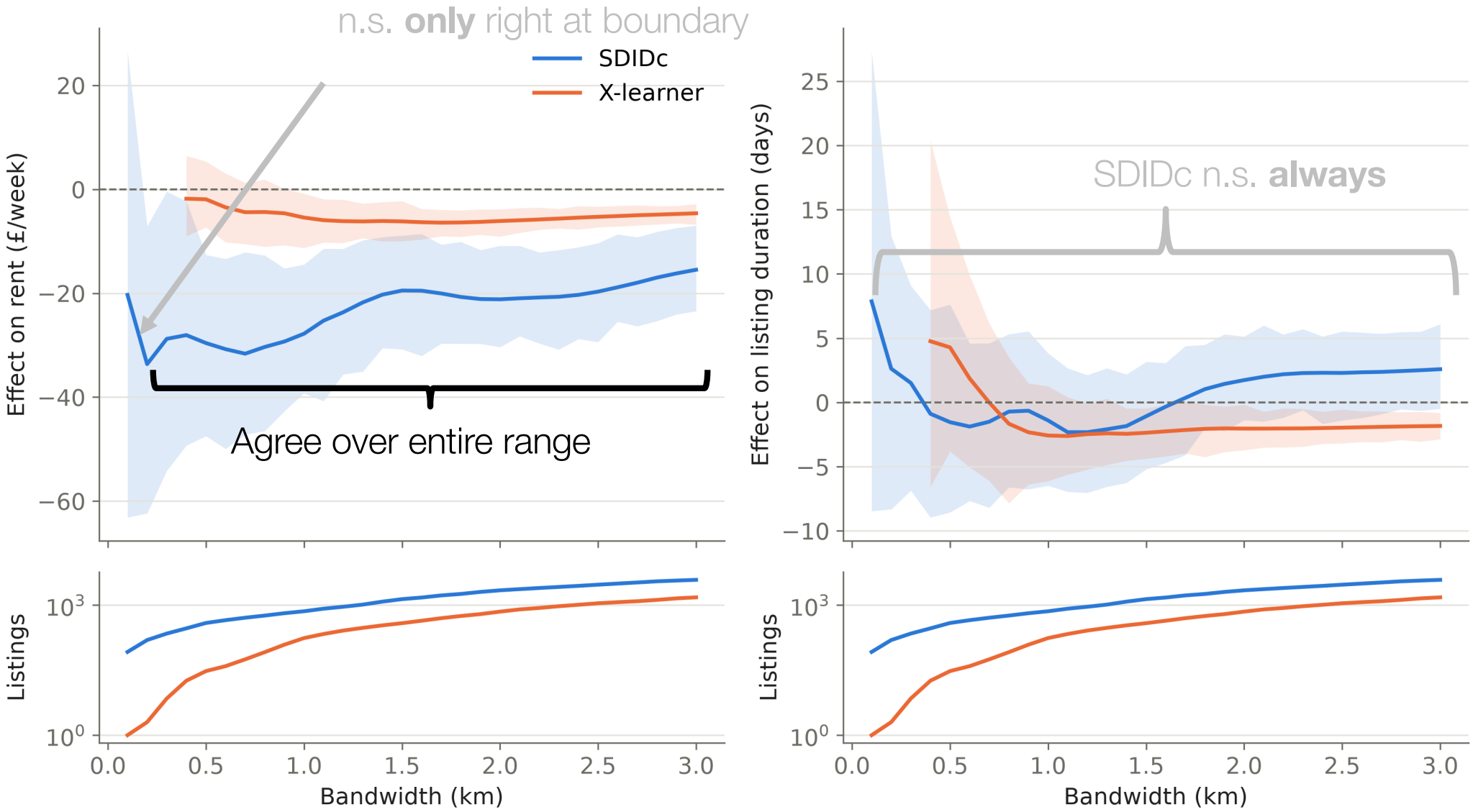
SDIDc VS XLEARNER BY BANDWIDTH



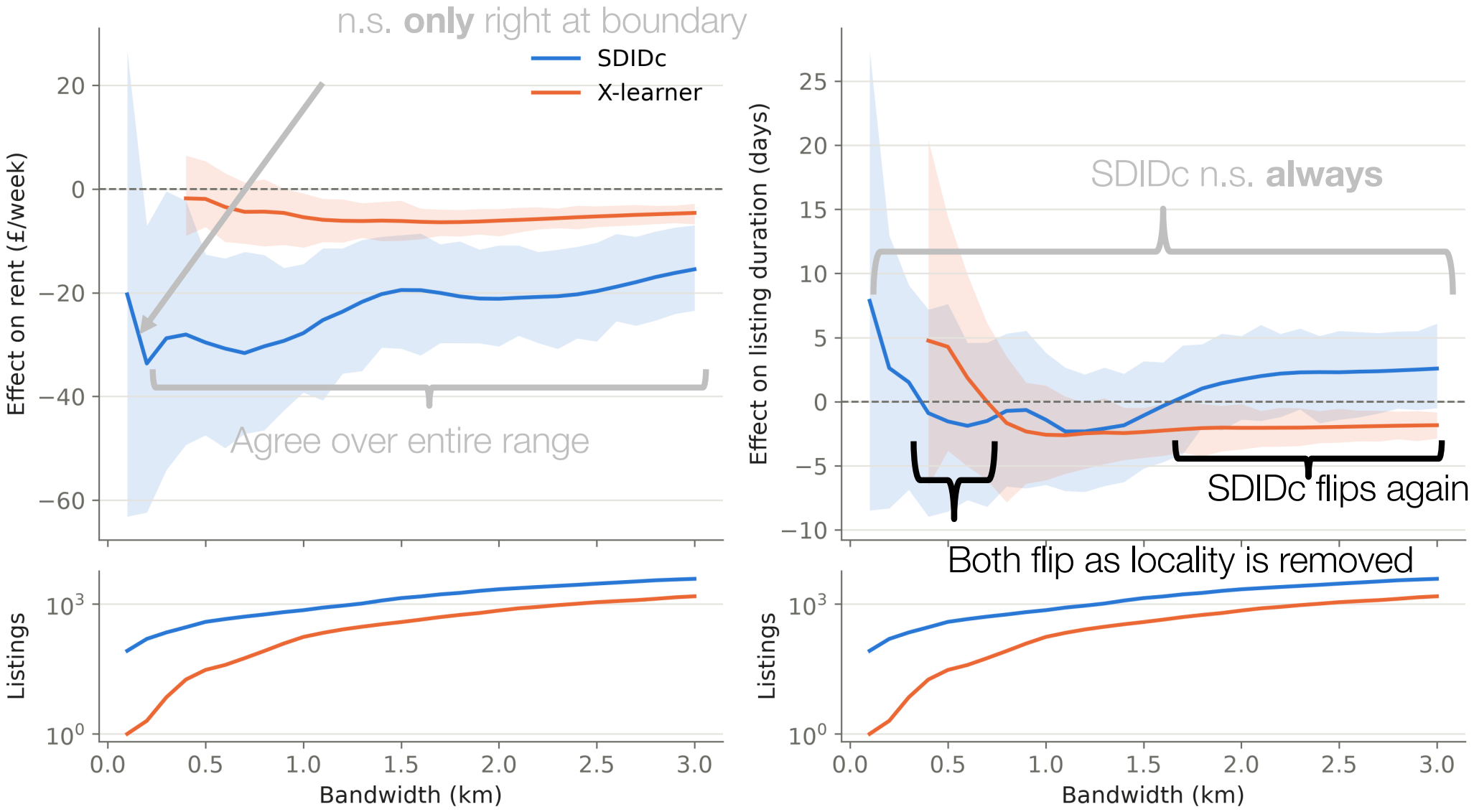
SDIDc VS XLEARNER BY BANDWIDTH



SDIDc VS XLEARNER BY BANDWIDTH



SDIDc VS XLEARNER BY BANDWIDTH



SDIDc VS XLEARNER BY BANDWIDTH

DLA rent premium narrowed between £20-£120, with no clear effect on clearing times, controlling for COVID-19, stock imbalance, path dependence.

Landlord concerns about fewer rentals & higher rents are not borne out in the short run.

Need access to property register to determine partial treatment, new rental market entrants

SLA OR NAY?

SLA OR NAY?

CAUSAL IMPACTS OF NEIGHBORHOOD-LEVEL RENT QUALITY REGULATION

Levi John Wolf & Alex Marsh
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